ASSIGNMENT -1 -CO1

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COURSE:	**MOBILE COMPUTING**

TASK 1 – Design Alternative Frequency-Planning Strategies

A fixed spectrum can be reused in different geographical areas. The same channels are assigned to cells that are sufficiently far apart so that their signals do not create unacceptable co-channel interference. Three distinct strategies are considered.

Strategy 1: Fixed Cluster Frequency Reuse (FCA)

The service area is divided into repeating clusters of N hexagonal cells. Each cell in a cluster receives a different channel group. The complete channel pattern repeats in the next cluster. For N=16, the 336 channels are divided into 16 groups of 21 channels each. This gives predictable planning and simple operation.

- Access-channel concept: in FDMA, each call receives a separate frequency pair; in TDMA, users share a carrier in different time slots; in OFDMA, users receive groups of subcarriers and resource blocks. Control channels carry access, paging and signalling; traffic channels carry voice/data.
- Assumptions: approximately regular hexagonal cells, planned antenna coverage, stable channel groups and a sufficiently uniform initial traffic map.
- Design principle: spatial reuse. The same channel group is reused only after the required reuse distance.
- Why investigate: it is simple, predictable and provides a baseline against which more flexible strategies can be compared.

Iterative pseudocode:

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INPUT: total_channels=336, candidate cluster sizes N, hexagonal grid\nFOR each valid N = i*i + i*j + j*j:\n compute Q = sqrt(3*N) and approximate SIR\n  IF SIR >= 12 dB:\n  divide 336 channels into N groups\n assign one group to each cell position in the cluster\n  repeat the cluster pattern over the grid\n calculate channels per cell and Erlang-B capacity\nOUTPUT: feasible N values and selected channel map
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Strategy 2: Sectorised Frequency Reuse

Each base station is divided into directional sectors, commonly three 120-degree sectors or six 60-degree sectors. Directional antennas concentrate energy towards the intended sector and reduce the number of strong co-channel interferers seen by a user. The same site can therefore reuse spectrum more efficiently, subject to antenna isolation and side-lobe leakage.

- Access channels are assigned to each sector. Control and traffic resources are planned per sector; the air-interface method may be FDMA/TDMA or OFDMA depending on the radio system.
- Assumptions: directional antenna patterns are installed correctly, sector boundaries are surveyed and neighbouring sectors are coordinated.
- Design principle: spatial filtering. Antenna direction reduces interference from unwanted directions.
- Why investigate: it can improve SIR without acquiring additional spectrum, especially in busy academic and stadium zones.

Pseudocode:

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FOR each base-station site:\n    divide the coverage area into 3 or 6 sectors\n    assign a coordinated channel/resource group to each sector\n    FOR each user location:\n        identify serving sector and strongest interfering sectors\n        estimate received wanted and interfering power using antenna gains\n        calculate SIR and blocking\n    adjust sector azimuth, tilt and power until coverage targets are met
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Strategy 3: Dynamic Channel Assignment and Borrowing (DCA)

Instead of permanently fixing every channel to one cell, a central controller or distributed radio controller assigns an available channel when a call arrives. A lightly loaded neighbouring cell may lend a channel to a busy cell, provided the borrowing does not violate interference constraints. When the call ends, the channel returns to the pool.

- Access channels are selected from a common or partially common pool. The controller checks availability, reuse distance, interference and handoff reservations before assigning a traffic channel.
- Assumptions: cells exchange load information, channel occupancy is updated quickly and a borrowing decision can be reversed safely.
- Design principle: trunking efficiency and adaptive resource allocation.
- Why investigate: traffic is not uniform. Hostels, lecture halls and stadiums can be busy at different times, so dynamic allocation can reduce blocking compared with rigid channel partitions.

Pseudocode:

```
WHEN a call request arrives in cell c:\n    list channels free in c and eligible neighbouring cells\n    reject channels that violate reuse/interference constraints\n    choose the eligible channel with the lowest predicted interference\n    if none is available, queue or block the request\n    otherwise assign channel and update occupancy\nWHEN call ends: release channel and update the pool
```

Preprocessing and cost responsibility

Before any strategy is selected, the operator needs a land-use and traffic map, building-height information, a propagation survey, candidate site locations, a hexagonal grid, road/access information and an estimate of busy-hour demand by zone. The operator normally bears the cost of radio planning, drive/walk testing, traffic surveys, engineering design and site acquisition. The university or township authority may assist with access to rooftops, poles and land, but the licensed operator remains responsible for network performance.

TASK 2 – Identify the Governing Design Parameter

Strategy	**Main governing parameter**	**Why it governs capacity/interference**	**Growth response**
Fixed cluster reuse	Cluster size N (and $Q=\sqrt{3N}$)	N controls channels per cell S/N and reuse distance; larger N improves SIR but reduces channels per cell.	Choose the smallest N that satisfies SIR; add cells or reduce R as demand rises.
Sectorisation	Number of sectors and effective interferer count	More directional sectors reduce strong interferers but increase sector boundaries and handoff complexity.	Increase sectorisation in concentrated high-load zones where antenna isolation is practical.
Dynamic assignment	Available channel pool and per-cell offered load A	The controller can use idle channels from other cells while preserving interference constraints.	More frequent load-aware reassignment and better coordination are needed at high demand.

For fixed reuse, N is the key trade-off. Increasing N increases reuse distance and SIR, but each cell receives fewer channels. For sectorisation, the effective number and strength of interferers are important. For dynamic assignment, the ratio of offered traffic to available channels in each cell is important.

Strategies and Governing Parameters

A.1 Background: the cellular concept

A cellular network divides a large service area into many small coverage areas called cells. Each cell is served by a base station (BS) with its own transmitter and receiver. Because the radio spectrum licensed to an operator is limited, the same frequencies must be used again in different cells. This is called frequency reuse. Reuse is possible because radio signals weaken with distance: if two cells that use the same channels are far enough apart, the interference between them is small enough to be tolerated. The ratio of the wanted signal power to the unwanted co-channel power is the signal-to-interference ratio (SIR), and the planner's job is to keep it above a minimum value (12 dB in this assignment) while giving each cell enough channels for its traffic.

The same idea explains the three strategies investigated in Task 1. Fixed cluster reuse controls interference by geometry (distance between co-channel cells). Sectorisation controls interference by direction (antenna patterns). Dynamic assignment controls interference by time (channels are allocated only when and where they are needed and when the interference check is passed). Each strategy therefore addresses the same two-sided problem: provide enough channels per cell and keep co-channel interference acceptable.

A.2 Key terms used in this assignment

Term	Meaning in this assignment
Cell / cluster	A cell is the area served by one BS (idealised as a hexagon). A cluster is a group of N cells that together use the whole set of channels once.
Cluster size N	Number of cells in one repeating cluster; $N=iZ+ij+jZ$. Here S=336 channels are divided among N cells.
Reuse distance D	Centre-to-centre distance between the nearest co-channel cells, $D=R\sqrt{(3N)}$.
Reuse ratio Q	$Q=D/R=\sqrt{(3N)}$. A larger Q means co-channel cells are further apart in cell radii.
SIR	Signal-to-interference ratio; the planning target is at least 12 dB.
Erlang (E)	Unit of traffic intensity. One Erlang is one channel occupied continuously for one hour. Each subscriber offers 0.025 E in the busy hour.
Blocking probability	Probability that a new call finds all channels busy; the design target is 2% (Erlang-B).
Handoff	Transfer of an active call from one cell to another as the user moves; more frequent in small cells.
Path-loss exponent n	Received power falls as distance^(−n); n≈2.5 to 4 for the campus and township environments considered.

A.3 Access methods behind the "access-channel" concept

Task 1 refers to access channels. The table below summarises how the three multiple-access techniques divide the same spectrum, because the unit that is reused or borrowed differs for each technique.

Technique	How resources are divided	Unit that is planned / reused	Typical use
FDMA	Separate frequency pair for each call	A frequency channel (carrier pair)	Analogue cellular, part of GSM planning
TDMA	Users share one carrier in different time slots	A carrier and time slot	GSM, digital cellular
OFDMA	Users are given groups of orthogonal subcarriers and time resource blocks	A resource block (subcarrier group × time)	LTE and 5G NR

Whatever the technique, the planning logic is identical: a set of radio resources is allocated to each cell or sector, the set is reused after the reuse distance, and control channels (access, paging, signalling) are protected so that calls can be set up even when traffic channels are busy.

A.4 Design inputs used throughout the assignment

Input	Value used	Where it is used
Total channels S	336	Divided into S/N channels per cell (Task 1, Task 3)
Planned cell radius R	0.5 km	Hexagon area 0.6495 kmZ, minimum 19 cells for 12 kmZ
Township area	12 kmZ	Coverage and capacity cell counts
Traffic per subscriber	0.025 E (25 mE)	Busy-hour offered traffic = subscribers × 0.025
Blocking target	2% (Erlang-B)	Maximum load per cell (14.036 E for 21 channels)
Minimum SIR	12 dB	Cluster-size feasibility check
Subscriber levels	1,000 / 10,000 / 20,000 / 100,000	Task 3, Task 5, Task 8
Path-loss exponent n	2.5, 3, 4	Task 3, Task 4, Task 5, Task 8

A.5 Comparison of the three strategies

Aspect	Fixed cluster reuse (FCA)	Sectorised reuse	Dynamic assignment (DCA)
Interference control	Geometry: reuse distance $D=R\sqrt{(3N)}$	Direction: antenna sectors reject interferers	Check at assignment time: only eligible channels are used
Capacity gain source	Smaller N or smaller cells	Lower SIR requirement allows smaller N	Pooling of idle channels (trunking efficiency)
Complexity	Low	Medium (antennas, azimuth, tilt, more handoffs)	High (controller, load exchange, reservations)
Best suited to	Uniform, predictable traffic	Concentrated, high-load zones	Non-uniform and time-varying traffic
Main weakness	Cannot follow traffic shifts	Side lobes and boundary handoffs	Needs fast signalling; limited by interference rules

Task 2 (continued) – how each parameter acts on the design

Task 2 identified one governing parameter for each strategy. The following notes explain the direction in which each parameter moves capacity and interference, so that the design trade-off can be read directly from the parameter.

- **Cluster size N (fixed reuse).** Increasing N lowers the number of channels per cell (S/N) and therefore lowers the Erlang-B capacity of the cell, but it raises $Q=\sqrt{(3N)}$ and the SIR. N is the only parameter that moves the two objectives in opposite directions, which is why it is the governing parameter.
- **Number of sectors (sectorisation).** Three 120° sectors or six 60° sectors reduce the number of strong first-tier interferers seen by a user. The cost is that the cell's channels are split between sectors, so trunking efficiency falls unless the sector load is balanced, and every sector boundary adds handoffs.

- **Offered load per cell A and the available pool (dynamic assignment).** The blocking probability depends on the ratio of offered traffic to channels. Dynamic assignment is valuable when this ratio is very different in neighbouring cells, because idle channels in one cell can serve the busy cell.

TASK 3 – Derive Capacity and Interference Functions Mathematically

3.1 Hexagonal geometry and cluster size

A regular hexagonal cell has six neighbouring cells. Moving between co-channel cells on a hexagonal grid by i cells in one direction and j cells at 60 degrees gives the cluster-size relation $N=i^2+ij+j^2$. Examples include N=3 (i=1,j=1), N=4 (2,0), N=7 (2,1), N=12 (2,2), and N=16 (4,0).

The centre-to-centre reuse distance is $D=R\sqrt{(3N)}$. Therefore $Q=D/R=\sqrt{(3N)}$. For N=16, $Q=\sqrt{48}=6.928$ and D=6.928R.

The area of a regular hexagon with centre-to-vertex radius R is $A_{cell}=(3\sqrt{3}/2)R^2=2.598R^2$. The number of cells required to cover area A is approximately C=ceil(A/A_cell), before adding overlap, edge coverage and reserve sites.

For R=0.5 km, $A_{cell}=2.598\times0.5^2=0.6495$ km². Thus the geometric minimum for 12 km² is ceil(12/0.6495)=19 cells. This is only a coverage estimate; capacity requires more cells.

3.2 Channels per cell and Erlang-B traffic capacity

For N=16, channels per cell m=S/N=336/16=21. Erlang-B is used because calls that arrive when all channels are occupied are blocked. Repeatedly applying $B(m,A)=A\cdot B(m-1,A)/(m+A\cdot B(m-1,A))$ gives a maximum offered load of approximately 14.036 Erlangs for 21 channels at 2% blocking. This value is an approximate planning limit per cell.

Subscribers	Total offered traffic	Approx. cells needed at 14.036 E/cell	Comment
1,000	25 Erlangs	2	Traffic is low; coverage and minimum site spacing dominate.
10,000	250 Erlangs	18	Capacity begins to require more cells than the initial coverage grid.
20,000	500 Erlangs	36	Additional capacity sites are required.
100,000	2,500 Erlangs	179	Dense small-cell deployment or other capacity enhancement is needed.

Calculation example for 100,000 subscribers: $A_{total}=100{,}000\times0.025=2{,}500$ Erlangs. Required cells=ceil(2,500/14.036)=179. The approximate cell radius needed to fit 179 equal hexagons into 12 km² is

TASK 3 (continued) – Mathematical Derivations

B.1 Proof of the cluster-size relation $N=i^2+ij+j^2$

In a hexagonal layout with cell radius R (centre to vertex), the distance between the centres of two adjacent cells is $\sqrt{3}\cdot R$. To locate the nearest co-channel cell, move i cells along one hexagon-edge direction, turn through 60° and move j cells in the new direction. The two displacement vectors have lengths $i\sqrt{3}R$ and $j\sqrt{3}R$, and the angle between them as vectors is 60°, so the triangle formed by the two moves and the straight line D has an interior angle of 120° opposite D. By the law of cosines:

$$D^2 = (i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R)\cos 120° = 3R^2(i^2 + j^2 + ij)$$

The area of the hexagon is $A_{cell} = 2.598RZ$, and a cluster of N cells has area $N\cdot 2.598RZ$. The co-channel cells form a lattice in which each co-channel cell owns one cluster, and the area per lattice point equals the area of the hexagon of circumradius $D/\sqrt{3}$ that surrounds it, namely $(3\sqrt{3}/2)(DZ/3) = 2.598\cdot DZ/3$. Equating this to $N\cdot 2.598RZ$ gives $DZ = 3N\cdot RZ$, hence $N = iZ+ij+jZ$ and $D = R\sqrt{(3N)}$. This confirms the relation used in Task 1 and Task 3.

B.2 Table of valid cluster sizes and reuse ratios

N	(i, j)	Q = √(3N)	D for R = 0.5 km (km)	Channels per cell (336/N)
1	(1, 0)	1.732	0.866	336.00
3	(1, 1)	3.000	1.500	112.00
4	(2, 0)	3.464	1.732	84.00
7	(2, 1)	4.583	2.291	48.00
9	(3, 0)	5.196	2.598	37.33 (not an integer)
12	(2, 2)	6.000	3.000	28.00
13	(3, 1)	6.245	3.122	25.85 (not an integer)
16	(4, 0)	6.928	3.464	21.00
19	(3, 2)	7.550	3.775	17.68 (not an integer)
21	(4, 1)	7.937	3.969	16.00
25	(5, 0)	8.660	4.330	13.44 (not an integer)
27	(3, 3)	9.000	4.500	12.44 (not an integer)
28	(4, 2)	9.165	4.583	12.00

The table shows why the cluster sizes N=3, 4, 7, 12 and 16 were chosen for testing: they all divide 336 exactly, so each cell receives a whole number of channels, whereas sizes such as N=9, 13, 19, 25 and 27 would leave fractional channel groups that must be rounded.

B.3 Derivation of the Erlang-B formula

Assume calls arrive as a Poisson process with offered load $A=\lambda/\mu$ Erlangs (λ is the arrival rate and $1/\mu$ the mean holding time), there are m identical channels, and blocked calls are cleared (not queued). The number of busy channels k is a birth–death process with arrival rate λ in every state $k<m$ and departure rate $k\mu$ in state k. The balance equations give $p_k = p_0\cdot A^k/k!$. Normalising so that the probabilities sum to one:

$$B(m, A) = p_m = (A^m/m!) / \Sigma_{k=0..m} (A^k/k!)$$

Direct evaluation of factorials is numerically unstable for m=21 or m=112, so the recursion $B(0,A)=1$, $B(m,A)=A\cdot B(m-1,A)/(m+A\cdot B(m-1,A))$ is used. It follows by dividing numerator and denominator of the formula for $B(m,A)$ by $\sum_{k=0..m-1} A^k/k!$ and substituting $B(m-1,A)$. This is the recursion applied in the Python function erlang_b(). The planning limit $A_{max}(m)$ is the largest A with $B(m,A)\leq 0.02$; because B increases monotonically with A, the program finds it by bisection (100 halvings of the interval 0–1000 E).

Check for m=21: B(21, 14.036) = 0.0200, i.e. 2.0%, and B(21, 15) = 0.0315, so the limit of 14.036 E quoted in Task 3 is confirmed.

B.4 Capacity per cell for different cluster sizes

N	Channels per cell m	Max load per cell at 2% blocking (E)	Load per channel (E/channel)	Trunking efficiency
3	112	99.624	0.889	88.9%
4	84	72.529	0.863	86.3%
7	48	38.392	0.800	80.0%
12	28	20.150	0.720	72.0%
16	21	14.036	0.668	66.8%
19	17	10.656	0.627	62.7%

The last column shows trunking efficiency: a cell with more channels carries more traffic per channel at the same blocking. N=3 gives 112 channels and 88.9% utilisation per channel, whereas N=16 gives 21 channels and 66.8%. This is the quantitative reason why larger clusters are less efficient even before the sites needed are counted.

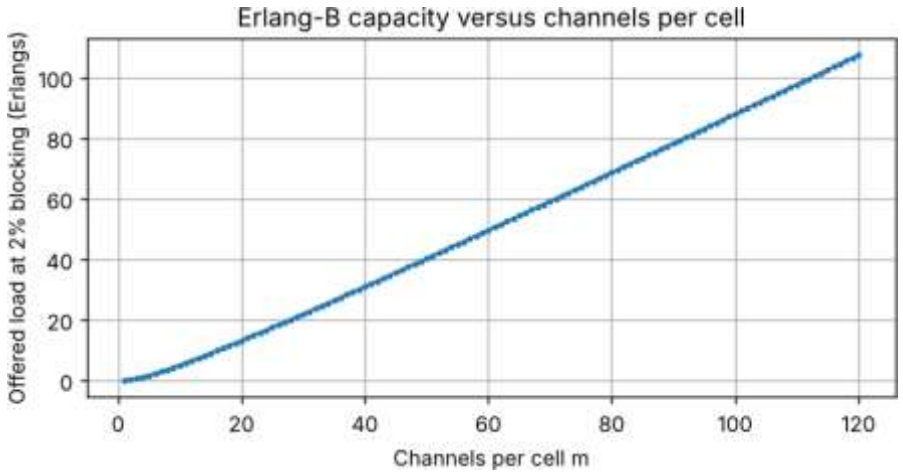


Figure S3.1 – Erlang-B capacity per cell at 2% blocking as a function of the number of channels (computed with the program of Task 8).



Figure S3.2 – Erlang-B blocking probability versus offered load for 14, 21 and 28 channels; the dashed line is the 2% target.

B.5 Worked example: cells needed for 10,000 subscribers with N=16

- Offered traffic: 10,000 × 0.025 = 250 E.
- Capacity per cell (21 channels, 2%): 14.036 E.
- Cells required: ceil(250/14.036) = ceil(17.81) = 18 cells.
- Coverage check: 18 cells × 0.6495 kmZ = 11.69 kmZ < 12 kmZ, so capacity and coverage are nearly balanced at this subscriber level; above this level capacity dominates the cell count, which agrees with the comment in the table of Section 3.2.

B.6 Extended SIR results for different cluster sizes and path-loss exponents

The general result SIR=$(3N)^{n/2}/6$ was evaluated for every tested combination. The values in dB are given below; entries below 12 dB are marked “fail”.

N	Q=√(3N)	n=2.5 (dB)	n=3 (dB)	n=4 (dB)
3	3.000	4.15 (fail)	6.53 (fail)	11.30 (fail)
4	3.464	5.71 (fail)	8.41 (fail)	13.80
7	4.583	8.75 (fail)	12.05	18.66
12	6.000	11.67 (fail)	15.56	23.34
16	6.928	13.23	17.44	25.84
19	7.550	14.17	18.56	27.34

From the table, the smallest valid cluster size that meets 12 dB under the six-equal-interferer approximation is N=13 for n=2.5 (12.1 dB), N=7 for n=3 (12.05 dB) and N=4 for n=4 (13.8 dB). The assignment adopts N=16 as a conservative planning case because the margin at N=13 is only about 0.1 dB, and shadowing of even 1 dB would remove it. The sanity check N=7, n=4 gives (21)Z/6 = 73.5 = 18.66 dB, which agrees with the stated 18.7 dB.

B.7 Sectorised SIR formula

If sectorisation leaves k_{eff} effective first-tier interferers instead of six, the same derivation gives SIR = $(3N)^{n/2}/k_{eff}$. With three 120° sectors the commonly used planning value is $k_{eff}=2$, and the simulator of Task 8 returns 18.01 dB for N=16 and n=2.5 instead of 13.23 dB, a gain of $10\log_{10}(6/2)=4.77$ dB. This gain is an upper estimate; with real side lobes the improvement is smaller, as the Monte-Carlo experiment in the Task 9 supplement shows.

B.8 Illustrative cost comparison (relative units)

The assignment does not give prices, so Section 3.4 leaves the cost model symbolic. To show how the model is used, the following table inserts *hypothetical relative cost units* (not real currency): C_survey=100, C_site=10 per site, C_sector=2 per site, C_control=40, and M=179 sites for 100,000 subscribers.

Strategy	C_total formula	C_total (units)	Capacity M·A_B (E)	Cost per Erlang (units/E)
Fixed reuse	C_survey + M·C_site	1890	2512	0.752
Sectorised	C_survey + M·C_site + M·C_sector	2248	2512 (same, before SIR gain is used)	0.895
Dynamic	C_survey + M·C_site + C_control	1930	2512 (same, before pooling gain is used)	0.768

These numbers only demonstrate the method. Sectorisation and dynamic assignment are justified when the capacity they release (a smaller M for the same traffic, or less blocking in hotspots) is worth more than the extra cost per site. A real study would replace the symbols with quotations for equipment, rent, power and backhaul.

B.9 Cells required for every cluster size and subscriber level

Combining the capacity per cell of Section B.4 with the traffic of each subscriber level (0.025 E per user) gives the capacity-limited cell count ceil(traffic / capacity per cell). The ideal equal-hexagon radius is $R=\sqrt{(12/(\text{cells}\times 2.598))}$ km.

Subscribers (traffic)	N=3	N=4	N=7	N=12	N=16
1,000 (25 E)	1	1	1	2	2
10,000 (250 E)	3	4	7	13	18
20,000 (500 E)	6	7	14	25	36
100,000 (2,500 E)	26	35	66	125	179

Subscribers	N=7: radius (km)	N=12: radius (km)	N=16: radius (km)
10,000	0.493	0.493	0.493
20,000	0.493	0.430	0.358
100,000	0.265	0.192	0.161

For the smaller subscriber levels the coverage minimum (19 cells at R=0.5 km) is larger than the capacity-limited count, so the radius used in the table is capped at the coverage value. Only when the capacity-limited count exceeds 19 does the radius have to shrink below 0.5 km. This is why 1,000 users need only the coverage grid, whereas 100,000 users need R≈0.16 km with N=16.

B.10 Why six interferers? Including the second and third tiers

The first-tier model uses six co-channel cells at distance D. In an infinite hexagonal pattern there are further interferers: six at about $\sqrt{3}\cdot D$ and six at 2D. Including them lowers the SIR slightly, because the extra power is small when n is large. The table compares the first-tier SIR with an 18-interferer calculation (wanted signal at distance R).

N	n	First tier only (dB)	Three tiers (dB)	Loss (dB)
7	2.5	8.75	7.19	1.55
7	3	12.05	10.85	1.20
7	4	18.66	17.97	0.70
12	2.5	11.67	10.12	1.55
12	3	15.56	14.37	1.20
12	4	23.34	22.65	0.70
16	2.5	13.23	11.68	1.55
16	3	17.44	16.24	1.20
16	4	25.84	25.15	0.70

The reduction is largest for small n (1.55 dB at n=2.5, 1.2 dB at n=3 and 0.7 dB at n=4 for N=16). The first-tier approximation is therefore slightly optimistic. For the open-campus case with n=2.5 the three-tier SIR for N=16 is 11.68 dB, about 0.3 dB below the 12 dB target, whereas the first-tier value is 13.23 dB. This does not invalidate the first-order design of the assignment, but it strengthens the conclusion of Task 11 that the open-campus case has little margin and that antenna optimisation and field measurements are necessary.

B.11 Coverage-limited cell count with allowances

Section 3.1 states that $C=\text{ceil}(A/A_{cell})$ is a minimum before overlap and reserve sites. If an allowance of 20% is added for overlap at the edge of the township and reserve sites, the planning count becomes ceil(1.2×18.48)=23 cells (18.48 is the unrounded ratio 12/0.6495). Using this planning count, the capacity of the coverage grid at N=16 is 23×14.036 = 322.8 E, supporting 12,914 subscribers at 0.025 E each. Beyond this number, additional capacity sites are needed.

B.12 Numerical illustration of the Erlang-B recursion

For a small example with A=2 E the recursion $B(m)=A\cdot B(m-1)/(m+A\cdot B(m-1))$ gives the following values, which can be checked by hand and show that blocking falls quickly as channels are added:

m	Calculation	B(m, 2)
0	B(0)=1	1.0000
1	2×1.0000/(1+2×1.0000)	0.6667
2	2×0.6667/(2+2×0.6667)	0.4000
3	2×0.4000/(3+2×0.4000)	0.2105
4	2×0.2105/(4+2×0.2105)	0.0952
5	2×0.0952/(5+2×0.0952)	0.0367
6	2×0.0367/(6+2×0.0367)	0.0121

The same procedure, repeated for m up to 21 with A=14.036, gives the 2% limit used throughout the assignment.

B.13 Trunking loss caused by splitting channels between sectors

Sectorisation divides the channels of a cell among its sectors. Because a smaller channel group has lower trunking efficiency, the total traffic that the three sectors can carry at 2% blocking is lower than that of the undivided cell, even though the SIR is better. The calculation below uses the 21 channels of the N=16 cell.

Layout	Channels per sector	Capacity per sector (E)	Total cell capacity (E)
Omnidirectional	21 (one group)	14.036	14.036
Three sectors, channels divided	7	2.935	8.806
Three sectors, each with 21 channels (new group per sector, smaller N)	21	14.036	42.108

If the 21 channels are simply divided into three groups of 7, the cell carries 8.81 E, about 37% less than the 14.04 E of the omnidirectional cell. Sectorisation therefore gives its capacity benefit only when the improved SIR is used to reduce the cluster size, so that each sector receives a channel group of the same size as before (last row). This is why Task 2 identifies the number of sectors and the effective interferer count, rather than the SIR alone, as the governing parameters.

R≈sqrt(12/(179×2.598))=0.161 km. This is a first-order estimate and does not include overlap, edge effects, uneven demand or reserve capacity.

3.3 Co-channel SIR derivation

Assume the wanted mobile is at distance R from its serving base station and the six first-tier co-channel base stations are each approximately D away. With path loss proportional to distance raised to n, wanted received power is proportional to R^(−n), while each interfering power is proportional to D^(−n). The total interference is approximately 6D^(−n). Hence SIR≈R^(−n)/[6D^(−n)]=(D/R)^n/6=(3N)^(n/2)/6.

For N=16: Q=√48=6.928. At n=2.5, SIR=48^1.25/6=21.06, or 13.23 dB. At n=3, SIR=48^1.5/6=55.43, or 17.44 dB. At n=4, SIR=48²/6=384, or 25.84 dB. The conservative open-campus estimate is therefore above the selected 12 dB minimum under the six-equal-interferer approximation.

Sectorisation changes the SIR denominator. If directional filtering reduces the number and power of dominant interferers, then SIR≈wanted power divided by the sum of the remaining weighted interferers. It is not correct to assume that sectorisation always divides the interferer count by three exactly; actual antenna patterns, side lobes and geometry must be measured.

Dynamic channel assignment does not automatically change the physical SIR formula. It selects only channels that pass the reuse/interference check. Its benefit is improved use of idle resources while maintaining the required SIR.

3.4 Planning cost and cost–capacity relation

Let C_site be the one-time cost of a site, C_survey the planning and measurement cost, C_sector the extra sector equipment cost, and C_control the controller/software cost. If M sites are installed, a simple planning-cost model is C_total=C_survey+M·C_site+M·C_sector+C_control. For fixed reuse, capacity is approximately M·A_B, where A_B is the Erlang-B traffic supported per cell. Thus cost per supported Erlang is C_total/(M·A_B).

FCA has lower control complexity but may need more sites when traffic is uneven. Sectorisation adds antenna and radio cost but may improve SIR and capacity per site. DCA adds controller and coordination cost but can reduce unused channel capacity in lightly loaded cells. Actual currency estimates cannot be calculated because the assignment does not provide equipment, rent, power or survey prices.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour

SIR depends on mobile position. Near the serving base station, the wanted signal is strong and SIR is normally best. At mid-cell, wanted power is lower and SIR is moderate. Near a cell edge or corner, the wanted signal is weaker and the mobile may be close to several neighbouring cells, so SIR is normally worst.

Case	Mobile position / traffic	SIR and blocking explanation

TASK 4 (continued) – Best, Average and Worst Cases

C.1 SIR as a function of mobile position

Section 3.3 evaluated the SIR for a mobile at the cell edge (distance R from the serving base station). For a mobile at distance d from its base station, with the six interferers still taken at distance D, the wanted power is proportional to d^{-n} and the SIR becomes $(D/d)^n/6$. Writing $d=x\cdot R$, where x is the fraction of the cell radius, $SIR(x) = (3N)^{n/2}/(6\cdot x^n)$. The table gives the idealised SIR for N=16 at four positions.

Position (x = d/R)	n=2.5 (dB)	n=3 (dB)	n=4 (dB)	Meets 12 dB at n=2.5?
0.25 (near BS)	28.29	35.50	49.93	Yes
0.50	20.76	26.47	37.88	Yes
0.75	16.36	21.19	30.84	Yes
1.00 (cell edge)	13.23	17.44	25.84	Yes

The values confirm the qualitative statement of Task 4: SIR is best near the base station and worst at the cell edge. At n=2.5 the SIR near the base station is more than 10 dB better than at the edge, so most of the cell has a comfortable margin and the edge governs the design. The improvement from x=1 to x=0.5 is $10\cdot n\cdot \log_{10}2 = 3.01\cdot n$ dB.

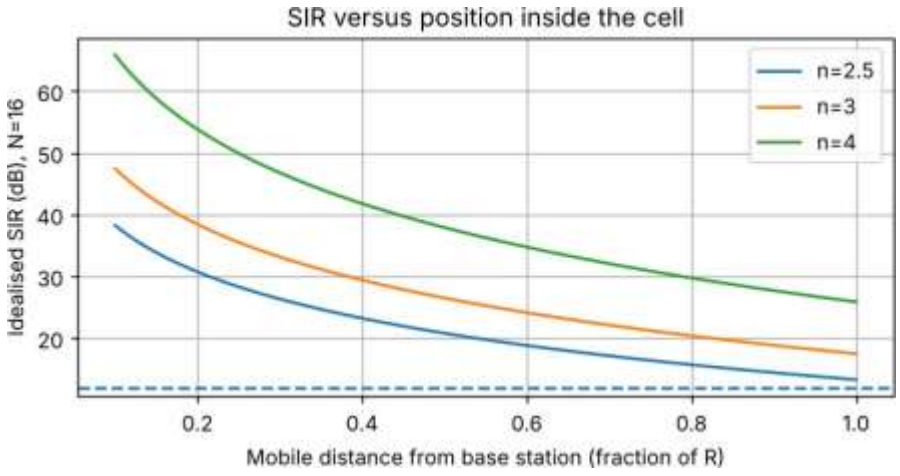


Figure S4.1 – Idealised SIR for N=16 versus distance from the serving base station for n=2.5, 3 and 4. The dashed line is the 12 dB minimum.

C.2 Effect of shadowing on the cell-edge margin

The cell-edge SIR for N=16 and n=2.5 is 13.23 dB, a margin of 1.23 dB above the 12 dB threshold. Any extra loss on the wanted signal (building shadowing, indoor penetration) reduces the margin one-for-one in dB if the interferers are unaffected. The table converts this to design guidance.

Extra loss on wanted signal (dB)	Edge SIR (dB)	Margin over 12 dB (dB)	Status
0	13.23	+1.23	Acceptable
1	12.23	+0.23	Acceptable
3	10.23	-1.77	Below target
6	7.23	-4.77	Below target
8	5.23	-6.77	Below target

A loss of only about 1.2 dB is enough to violate the target at the edge when n=2.5, which is why the assignment states that the design needs field validation. At n=3 the margin is 5.44 dB and at n=4 it is 13.84 dB, so shadowing is a concern mainly in the open-campus case.

C.3 Blocking in the best, average and worst traffic cases

Blocking depends on the offered load relative to the 21 channels of a cell. The Erlang-B values below illustrate the three traffic cases of Task 4.

Offered load A (E)	Load / capacity (14.036 E)	Blocking B(21, A)	Interpretation
5	36%	0.00%	Best case: off-peak
8	57%	0.01%	Light busy-hour load
10	71%	0.09%	Moderate
12	85%	0.56%	Average busy hour, heavily used cell
14.036	100%	2.00%	Planned limit (design point)
16	114%	4.68%	Overloaded: event surge begins
20	142%	13.14%	Worst case: severe surge
25	178%	24.99%	Extreme surge

Blocking rises from almost zero to about 2% at the design point, then increases very quickly. At 20 E (about 1.4 times the design load) roughly 13% of calls, about one in eight, are blocked. This is the reason the worst case in Task 4 is described as a sharp rise rather than a gradual one, and it connects directly to the stadium event of Task 12.

C.4 Summary of the three cases

Case	Radio condition (N=16, n=2.5)	Traffic condition	Expected result
Best	x=0.25: SIR ≈ 28.3 dB	5 E (36% load): blocking ≈ 0.0%	High quality, no blocking
Average	x≈0.6–0.75: SIR ≈ 16–19 dB	10–12 E: blocking ≈ 0.1–0.6%	Normal service
Worst	x=1 with 6 dB shadowing: SIR ≈ 7.2 dB	20 E: blocking ≈ 13%	Outage at the edge plus blocking

C.5 Fraction of the cell area that meets the 12 dB target

Since SIR falls as the mobile moves outwards, there is a limiting distance x_{max} beyond which the SIR drops below 12 dB. Setting SIR(x)=12 dB gives $x_{max} = 10^{(SIR_{edge} - 12)/(10n)}$. If $x_{max} \geq 1$ the whole cell meets the target; otherwise the fraction of the (circular) cell area that meets it is x_{max}^{Z}.

N	n	Edge SIR (dB)	x_{max}	Fraction of area meeting 12 dB
7	2.5	8.75	0.74	55%
7	3	12.05	≥ 1.00	100%
7	4	18.66	≥ 1.00	100%
12	2.5	11.67	0.97	94%
12	3	15.56	≥ 1.00	100%
16	2.5	13.23	≥ 1.00	100%
16	3	17.44	≥ 1.00	100%

The table shows that N=7 at n=2.5 meets the target over only about 55% of the cell area, N=12 at n=2.5 over about 94%, and N=16 over the entire cell in the idealised model. These ideal fractions ignore shadowing; in practice the coverage fraction is lower, which is why a margin is added.

C.6 Causes of the worst case and countermeasures

Worst-case cause	Effect on SIR / blocking	Countermeasure
Mobile at cell edge or corner	Lowest wanted power, several equal interferers	Sectorisation, antenna tilt, handoff margin tuning
Deep shadowing behind buildings	Wanted signal falls by several dB	Additional micro/pico cells; indoor coverage
Event surge or hotspot	Load above 14.036 E, blocking rises quickly	Dynamic borrowing, temporary cells, cell splitting
Synchronised busy hour	Many cells loaded together; no spare channels to borrow	Capacity growth by splitting; load-aware planning
Poor reuse planning	Co-channel cells closer than D	Re-check reuse distance after every site change

In an academic township the lecture breaks, hostel evenings and stadium events do not coincide, so a plan based only on the network-wide average would miss the localised worst case. This is the key reason why Task 4 treats position and traffic together.

Best	Near serving base station; off-peak	Short wanted-signal distance gives high received power. Few channels are occupied, so blocking is close to zero.
Average	Typical locations; average busy hour	Wanted and interfering powers are moderate. Erlang-B blocking approaches the design target only when the cell is loaded near its planned limit.
Worst	Cell edge/corner; event surge	Wanted signal is weak and multiple neighbours may interfere. High offered load fills channels and blocking rises sharply.

Example: for N=16 and n=2.5, the ideal six-interferer SIR estimate is 13.23 dB. This is a cell-edge planning approximation. Near the base station the wanted distance is smaller than R, so SIR increases. If shadowing reduces the wanted signal by 6 dB, the effective SIR may also fall by roughly 6 dB; therefore a design with only 1.23 dB margin above the 12 dB threshold needs field validation.

The busy-hour offered traffic for 1,000 subscribers is 25 E, for 10,000 it is 250 E, for 20,000 it is 500 E and for 100,000 it is 2,500 E. Blocking is not determined by subscriber count alone; it depends on how traffic is distributed among cells and the number of channels in each cell. A local event can overload one zone even when the township-wide average is acceptable.

A larger path-loss exponent reduces distant co-channel interference more rapidly, improving the ideal SIR estimate. However, dense buildings can also create shadowing, penetration loss and coverage holes. Therefore n=4 does not automatically mean better user experience; it can mean weaker coverage behind obstacles.

TASK 5 – Scaling and Sensitivity Analysis

For fixed reuse, channels per cell are S/N, so the number of channels available per cell is proportional to 1/N. Under the six-interferer approximation, SIR is proportional to N^(n/2). For fixed township area, the number of cells is proportional to $1/R^2$. Offered traffic is proportional to the number of subscribers.

Change	**Mathematical effect**	**Planning interpretation**
N increases	Channels/cell $\propto$ 1/N; SIR $\propto$ N^(n/2)	Better interference protection but fewer channels in each cell.

TASK 5 (continued) – Scaling and Sensitivity

D.1 Sensitivity to the cluster size N

The table combines the three effects of Task 5 for each tested cluster size: channels per cell, traffic capacity per cell and SIR. The last columns give the number of cells needed for 250 E (10,000 subscribers) and 2,500 E (100,000 subscribers) from the Task 8 program.

N	Channels/cell	Capacity/cell (E)	SIR n=2.5 (dB)	SIR n=4 (dB)	Cells for 250 E	Cells for 2,500 E
3	112	99.62	4.15	11.30	3	26
4	84	72.53	5.71	13.80	4	35
7	48	38.39	8.75	18.66	7	66
12	28	20.15	11.67	23.34	13	125
16	21	14.04	13.23	25.84	18	179

Moving from N=7 to N=16 more than doubles the number of cells needed (66 to 179 for 100,000 subscribers) while the SIR at n=2.5 improves by only 4.5 dB. The relation capacity per cell 1/N is not exactly linear, because the Erlang-B limit grows slightly faster than the channel count; from N=16 to N=7 the channels per cell increase 2.29 times while the capacity per cell increases 2.73 times.

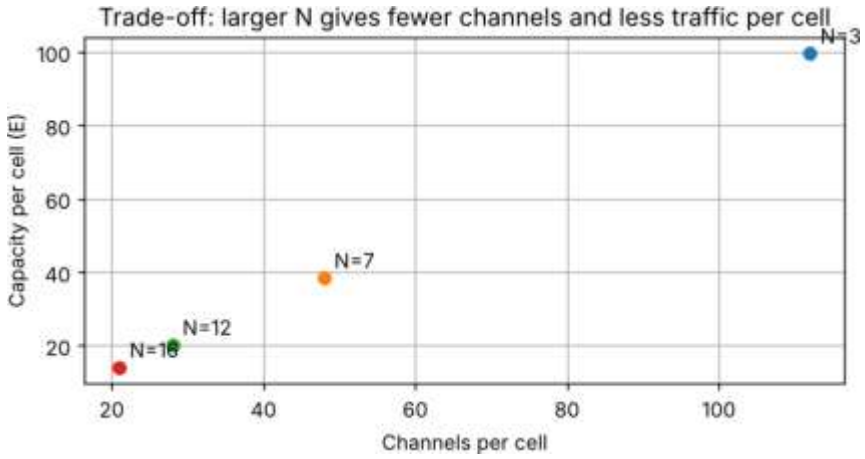


Figure S5.1 – Capacity per cell and channels per cell for the tested cluster sizes.

D.2 Sensitivity to the cell radius R

Because the number of cells for a fixed area scales as 1/RZ, halving the radius quadruples the cell count. The following table gives the idealised hexagon area and the geometric number of cells for the 12 kmZ township.

R (km)	Cell area $2.598R^2$ (km²)	Cells for 12 km² (geometric)	Cell count relative to R=0.5 km
0.5	0.6495	19	1.0×
0.4	0.4157	29	1.5×
0.3	0.2338	52	2.7×
0.25	0.1624	74	3.9×
0.2	0.1039	116	6.1×
0.161	0.0673	179	9.4×

The 179 cells needed for 100,000 subscribers correspond to R≈0.161 km, which is about 3.1 times smaller than the planned 0.5 km and needs about 9.4 times as many cells as the coverage minimum of 19. This explains the recommendation to move to a hierarchical macro–micro design.

D.3 Sensitivity to the number of subscribers

Subscribers	Offered traffic (E)	Cells at N=16	Cells at N=7	Cells at N=3	Average load per cell at N=16 (E)
1,000	25	2	1	1	12.50
10,000	250	18	7	3	13.89
20,000	500	36	14	6	13.89
50,000	1,250	90	33	13	13.89
100,000	2,500	179	66	26	13.97

Offered traffic is linear in the subscriber count, so the cell count is also linear when the cell capacity is fixed: doubling the population doubles the sites unless capacity per site is raised by sectorisation, borrowing or smaller cells. The 50,000-subscriber row is added only to show the linear trend between the 20,000 and 100,000 cases of the assignment.

D.4 Sensitivity ranking

Comparing the changes, the capacity needed for a given population is most sensitive to the cell radius (an inverse-square effect) and to the cluster size N (an inverse effect on channels per cell), whereas the path-loss exponent changes the interference margin but not the Erlang-B capacity. In practice the planner should therefore (1) choose the smallest N that satisfies the SIR target with a margin, (2) reduce R only where the traffic map shows demand, and (3) treat n as an uncertainty to be measured in a propagation survey.

Parameter changed	Effect on capacity	Effect on SIR	Main side effect
N doubled	Channels per cell roughly halved	Increases by (2)^(n/2); +3.8 dB at n=2.5	Fewer channels; more cells needed
R halved	Cell count × 4	Unchanged for the same N (ratio D/R)	Handoffs × 2, site cost up
Subscribers × 2	Traffic × 2	Unchanged	More sites or sectors required
n from 2.5 to 4	Unchanged	+12.6 dB at N=16 (idealised)	More shadowing and coverage holes

D.5 Break-even capacity of a given number of cells

The number of subscribers that a layout can support before the Erlang-B limit is reached is subscribers = (cells × capacity per cell)/0.025. This gives the cell-splitting trigger point for each cluster size.

Cells	N=7 (38.39 E per cell)	N=12 (20.15 E per cell)	N=16 (14.036 E per cell)
19	29,178	15,314	10,667
25	38,392	20,150	14,036
50	76,783	40,301	28,072
100	153,567	80,602	56,144
179	274,884	144,277	100,498

For example, the 19-cell coverage grid at N=16 supports about 10,667 subscribers before blocking exceeds 2% in the average cell. This is the break-even point that triggers the first capacity actions of Task 5. In practice, because traffic is not uniform, individual cells reach their limit earlier than the average, so the trigger should be based on the busiest cells.

D.6 Growth plan derived from the sensitivity results

Stage	Subscribers	Main action	Reason
1	Up to about 10,000	Coverage grid of 19 cells, N=16, fixed reuse	Coverage dominates; simplest operation
2	10,000 to 20,000	Sectorise busy sites; split hostel and academic-block cells; enable borrowing	Capacity begins to dominate; local hotspots appear
3	20,000 to 100,000	Micro/small cells in dense zones; macro layer for coverage	Cell radius must approach 0.16 km on average
4	Above 100,000	Continued densification and capacity optimisation	A macro-only layout is not economical

R decreases	Cell count $\propto 1/R^2$	More sites and higher aggregate capacity, but more handoffs and site cost.
Subscribers increase	Traffic $\propto$ subscribers	Capacity must rise through smaller cells, sectorisation and/or dynamic assignment.
n rises from 2.5 to 4	Ideal SIR increases for a fixed N	Distant interference falls faster, but buildings may cause shadowing and coverage loss.

The break-even point for cell splitting is reached when a cell's offered traffic approaches its Erlang-B limit. With 21 channels and a 2% target, that limit is about 14.036 E. If the cell carries more traffic, the operator should add capacity by splitting, sectorising or borrowing channels. Above 100,000 subscribers, a macro-only design is unlikely to be economical; a hierarchical macro–micro/small-cell layout is more practical.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting

A cell is split by reducing its radius from R to R/2. Since cell area is proportional to R^2, each new cell has one-quarter of the original area. One parent cell is replaced by four child cells. If all cells are split uniformly, the total number of cells and ideal aggregate capacity each increase by a factor of four per round, assuming the same channel reuse and a similar traffic distribution.

Let C_k be the number of cells after k rounds, K_k be the ideal aggregate offered-traffic capacity, P_k be the transmit power per cell, and H_k be the relative handoff rate. Base case: C_0 is the original cell count, K_0 is its aggregate capacity, P_0 is the original transmit power and H_0 is the original handoff rate.

Quantity	**Uniform-splitting recurrence**	**Closed form**
Cell count	C_k = 4C_(k−1)	C_k = C_0·4^k
Aggregate capacity	K_k = 4K_(k−1)	K_k = K_0·4^k
Cell radius	R_k = R_(k−1)/2	R_k = R_0/2^k
Transmit power	P_k = P_(k−1)/2^n	P_k = P_0/2^(nk)
Relative handoff rate	H_k ≈ 2H_(k−1)	H_k ≈ H_0·2^k

The power relation follows the assignment's model $P' = P/2^n$: halving distance reduces the power needed to maintain the same received level under path-loss exponent n. Actual power control is limited by minimum transmit power, interference, indoor penetration and regulatory limits.

TASK 6 (continued) – Recurrence for Cell Splitting

E.1 Derivation of the uniform-splitting recurrence

A regular hexagonal cell of radius R has area 2.598RZ. Splitting reduces the radius to R/2, so the area of each new cell is 2.598(R/2)Z = 0.25 of the original. Covering the same area therefore needs four new cells per old cell, which gives $C_k = 4C_{k-1}$. Each new cell keeps the same set of channels (the cluster size N is unchanged, so the reuse ratio D/R is unchanged), so the channels per unit area increase four times and the aggregate traffic capacity satisfies $K_k = 4K_{k-1}$. Unrolling the recurrence k times gives the closed form $C_k = C_0 \cdot 4^k$, which can be verified by induction: $C_0 \cdot 4^0 = C_0$ and, if $C_{k-1} = C_0 \, 4^{k-1}$, then $C_k = 4C_{k-1} = C_0 \, 4^k$.

E.2 Derivation of the transmit-power relation

To keep the same received power at the cell edge, the transmit power must compensate path loss: $P_r = P \cdot R^{-n}$. After splitting, $P' \cdot (R/2)^{-n} = P \cdot R^{-n}$, so $P' = P/2^n$. After k rounds, $P_k = P_0/2^{nk}$. For n=2.5 one round reduces power by $2^{2.5} = 5.66$ times (7.5 dB); for n=3 by 8 times (9 dB); for n=4 by 16 times (12 dB).

E.3 Numerical table for repeated uniform splitting

The table uses $R_0 = 500$ m, $K_0 = 14.036$ E per original cell and $C_0 = 1$ (all values are per original cell).

k	Cells C_k/C_0	Radius R_k (m)	Capacity K_k (E)	Power P_k/P_0 (n=2.5)	Power P_k/P_0 (n=4)	Handoff rate H_k/H_0
0	1	500.0	14.0	1.00000	1.0000000	1
1	4	250.0	56.1	0.17678	0.0625000	2
2	16	125.0	224.6	0.03125	0.0039062	4
3	64	62.5	898.3	0.00552	0.0002441	8
4	256	31.2	3,593.2	0.00098	1.53e-05	16
5	1024	15.6	14,372.9	0.00017	9.54e-07	32

The table shows the key trade-off of the recurrence: capacity grows by 4 at every round while the handoff rate only doubles, so splitting is effective, but the number of sites grows with the capacity. After k=3 the radius is 62.5 m, which is the order of the 65 m cell size needed in the stadium zone in Task 12.

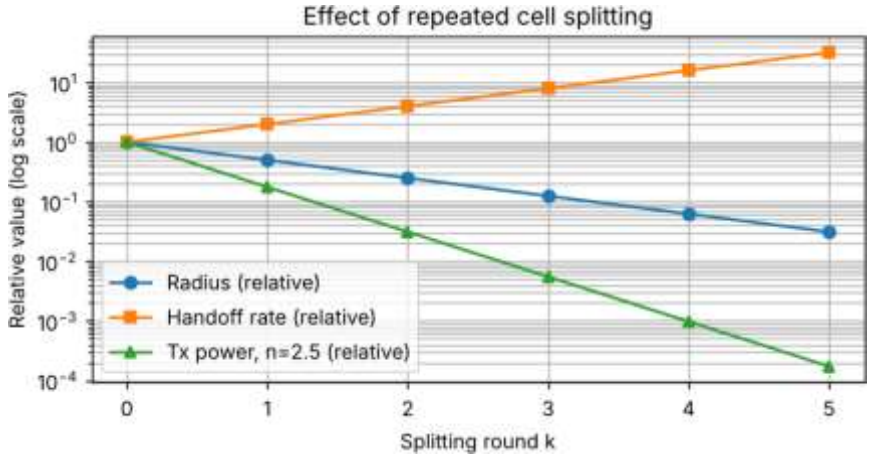


Figure S6.1 – Relative radius, handoff rate and transmit power (n=2.5) after k rounds of uniform splitting.

E.4 Selective splitting: cells added per round

When a fraction f of the cells is overloaded, $C_k = C_{k-1}(1+3f)$ because each split cell is replaced by four, a net gain of three. Starting from $C_0 = 19$ cells, the table gives the cell counts for different values of f.

k	f=0.1 (×1.3)	f=0.25 (×1.75)	f=0.5 (×2.5)	f=1.0 (×4, uniform)
0	19.0	19.0	19.0	19.0
1	24.7	33.2	47.5	76.0
2	32.1	58.2	118.8	304.0
3	41.7	101.8	296.9	1216.0
4	54.3	178.2	742.2	4864.0

Values are not integers because the model is an average; in practice the number of cells split is rounded to a whole number. For f=1 the model reduces to uniform splitting (19, 76, 304, 1,216 …). A small f gives a slow, targeted growth that suits a township in which only the hostels, academic blocks and the stadium are overloaded.

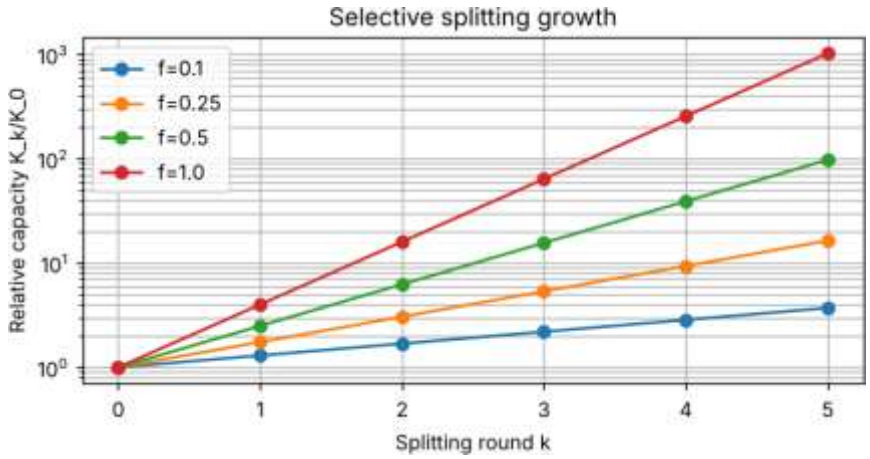


Figure S6.2 – Relative capacity growth K_k/K_0 for selective splitting with different split fractions f.

E.5 Limits of the recurrence model

- Transmit power cannot fall without limit: minimum power, receiver sensitivity and indoor penetration set a floor.
- Handoff rate growth of 2^k assumes constant user speed; fast users cross small cells even more often.
- Capacity growth of 4^k assumes the same traffic density and the same reuse; the interference level and backhaul requirement increase with the site count.

E.6 Worked example: township with 19 initial cells

Take $C_0 = 19$ cells, $K_0 = 19 \times 14.036 = 266.7$ E and a forecast demand D=2,500 E (100,000 subscribers). Then $D/K_0 = 9.37$ and $k_{min}^0 = \text{ceil}(\log_4 9.37) = \text{ceil}(1.62) = 2$. After two rounds of uniform splitting the township would have $19 \times 16 = 304$ cells and an ideal capacity of 4,267 E, which is 1.71 times the demand. In Task 3 only 179 cells were needed, so uniform splitting overshoots by 304−179=125 cells (70% more than necessary).

Approach	Cells	Capacity (E)	Comment
No splitting (19 cells)	19	266.7	Fails: demand 2,500 E
Uniform splitting, k=1	76	1,066.7	Fails
Uniform splitting, k=2	304	4,266.8	Meets demand with 71% surplus
Capacity-matched layout (Task 3)	179	2,512.4	Meets demand; requires non-uniform splitting

The comparison shows why the selective recurrence $C_k = C_{k-1}(1+3f)$ is more realistic: a smaller number of targeted splits reaches the required capacity with fewer sites. With f=0.5 the model gives 19×2.5^k: 47.5 cells after one round, 118.8 after two and 296.9 after three, so about 2.5 rounds would reach 179 cells.

E.7 Transmit-power reduction in decibels

k	n=2.5: reduction (dB)	n=3: reduction (dB)	n=4: reduction (dB)
1	7.5	9.0	12.0
2	15.1	18.1	24.1
3	22.6	27.1	36.1
4	30.1	36.1	48.2

Each round lowers the required power by 3.01·n dB. This shows that the base-station power is not a limiting factor for dense small cells; the real limits are backhaul, site acquisition and handoff management.

Selective splitting: if only a fraction f of the current cells is overloaded, each selected cell is replaced by four cells while the remaining cells are retained. Therefore $C_k=C_{(k-1)}(1+3f)$. Under the simplifying assumption that each split cell adds three extra cell-capacity units, $K_k=K_{(k-1)}(1+3f)$. If the fraction varies by round, $C_k=C_0\prod(1+3f_j)$. This selective model is more realistic for a township with concentrated demand.

TASK 7 – Solve the Recurrence and Interpret the Result

For uniform splitting, $K_k=K_0\cdot 4^k$. To support forecast demand D, require $K_0\cdot 4^k \geq D$. Dividing by K_0 gives $4^k \geq D/K_0$. Taking logarithms gives $k \geq \log_4(D/K_0)$. Since k must be a whole number, the minimum number of rounds is $k_{min}=\text{ceil}(\log_4(D/K_0))$.

Example: if one original cell supports $K_0=14.036$ E and the required traffic is D=200 E, then $k_{min}=\text{ceil}(\log_4(200/14.036))=\text{ceil}(1.91)=2$ rounds. The ideal capacity becomes $14.036\times16=224.576$ E. The number of cells also becomes 16 times the original number, the radius becomes $R_0/4$, transmit power per cell becomes $P_0/4^n$, and the relative handoff rate becomes about four times the original.

For selective splitting with a constant split fraction f, $K_k=K_0(1+3f)^k$. Thus $k_{min}=\text{ceil}(\log(D/K_0)/\log(1+3f))$. This formula applies when $0<f\leq1$ and the simplified capacity scaling assumption is reasonable. If f=0, no capacity growth occurs through splitting.

Splitting increases capacity but also increases site count, backhaul demand, handoff frequency, maintenance and interference coordination. Therefore the mathematically smallest k is not automatically the economically best design.

TASK 8 – Implement and Test the Planning Strategies

The following compact Python program provides a reproducible analytical simulator. It calculates valid cluster sizes, channels per cell, approximate SIR, Erlang-B traffic limits, traffic demand and the cell count required. It compares fixed reuse, a sectorisation approximation and dynamic channel pooling. The sector and dynamic results are simplified planning indicators; a full Monte-Carlo radio simulation requires real antenna patterns, user coordinates, shadowing maps and call-arrival traces.

```
import math

TOTAL_CHANNELS = 336
BLOCKING_TARGET = 0.02
TRAFFIC_PER_USER = 0.025
MIN_SIR_DB = 12.0
TOWNSHIP_AREA = 12.0
```

```
def valid_cluster_sizes(limit=30):
    values = set()
    for i in range(limit):
        for j in range(limit):
            n = i*i + i*j + j*j
            if 1 <= n <= limit:
                values.add(n)
    return sorted(values)

def sir_db(N, path_loss_n, interferers=6):
    sir = (3*N)**(path_loss_n/2) / interferers
    return 10*math.log10(sir)

def erlang_b(channels, offered):
    b = 1.0
    for m in range(1, channels + 1):
        b = offered*b/(m + offered*b)
    return b

def max_erlangs(channels, target=0.02):
    low, high = 0.0, 1000.0
    for _ in range(100):
        mid = (low + high)/2
        if erlang_b(channels, mid) <= target:
            low = mid
        else:
            high = mid
    return low

def evaluate_fixed(N, n=2.5):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    return channels, capacity, sir_db(N, n)

def evaluate_sectorised(N, n=2.5, effective_interferers=2):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
```

```
    sir = sir_db(N, n, effective_interferers)
    return channels, capacity, sir

def evaluate_dynamic(N, n=2.5):
    # Dynamic borrowing uses the same physical SIR eligibility check.
    # Pooling can improve utilisation, but does not create new spectrum.
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    sir = sir_db(N, n)
    return channels, capacity, sir

def main():
    print("Valid cluster sizes:", valid_cluster_sizes())
    for users in [1000, 10000, 20000, 100000]:
        traffic = users * TRAFFIC_PER_USER
        print("\nSubscribers:", users, "Traffic:", traffic, "Erlangs")
        for N in [3, 4, 7, 12, 16]:
            m, cap, sir = evaluate_fixed(N, 2.5)
            cells = math.ceil(traffic/cap)
            print("N", N, "channels/cell", m,
                  "capacity/cell", round(cap, 2),
                  "SIR dB", round(sir, 2),
                  "cells", cells)
    print("\nN=16 sectorised illustration:",
          evaluate_sectorised(16, 2.5, 2))
    print("N=16 dynamic assignment baseline:",
          evaluate_dynamic(16, 2.5))

if __name__ == "__main__":
    main()
```

How to run: install Python 3, save the code as cellular_planning.py, open a terminal in the saved folder and run: python cellular_planning.py. The same inputs must be used for all strategies. For a full Monte-Carlo extension, generate random user positions in each hexagon, calculate path loss and log-normal shadowing, apply antenna gain by sector, select eligible channels and record SIR for each trial. Use a fixed random seed so that every strategy is tested with the same user positions.

TASK 7 (continued) – Solving the Recurrence

F.1 Minimum number of splitting rounds for different demands

Using $k_{min} = \text{ceil}(\log_4(D/K_0))$ with $K_0 = 14.036$ E (one original cell), the following table gives the number of rounds needed for several demand levels. The logarithm is computed as $\ln(D/K_0)/\ln 4$.

Demand D (E)	D/K_0	$\log_4(D/K_0)$	k_{min}	Capacity K_k (E)	Cells (× original)
20	1.42	0.26	1	56.1	4
50	3.56	0.92	1	56.1	4
100	7.12	1.42	2	224.6	16
200	14.25	1.92	2	224.6	16
500	35.62	2.58	3	898.3	64
1,250	89.06	3.24	4	3,593.2	256
2,500	178.11	3.74	4	3,593.2	256

The row D=200 E reproduces the example in Task 7 (1.91, so $k_{min} = 2$, capacity 224.576 E). The row D=1,250 E is the stadium event of Task 12; for one original cell it would need k=4 rounds (256 times the original cells), which shows why uniform splitting of the whole township would be wasteful for a temporary, localised event.

F.2 Selective splitting: rounds needed for a constant fraction f

For D=200 E and $K_0 = 14.036$ E, $D/K_0 = 14.25$. Applying $k_{min} = \text{ceil}(\ln(14.25)/\ln(1+3f))$:

f	1+3f	ln(1+3f)	k_{min} = ceil(2.657 / ln(1+3f))
0.1	1.30	0.262	11
0.25	1.75	0.560	5
0.5	2.50	0.916	3
1.0	4.00	1.386	2

A smaller fraction needs more rounds (f=0.1 needs 11 rounds, f=1 only 2) but splits fewer cells in each round. The planner must therefore decide between a few large expansion programmes and many small, targeted ones; the economic comparison depends on site cost and construction lead time, which the assignment does not specify.

TASK 8 (continued) – Implementation and Testing

F.3 Explanation of the program

Function	Purpose	Formula / method
valid_cluster_sizes()	Lists feasible N	N=iZ+ij+jZ for i, j in 0…29 with N ≤ 30
sir_db(N, n, interferers)	Approximate co-channel SIR in dB	10·log10((3N)^(n/2)/k)
erlang_b(channels, offered)	Blocking probability	Recursion B=A·B/(m+A·B)
max_erlangs(channels, target)	Largest load with blocking ≤ 2%	Bisection on A between 0 and 1000
evaluate_fixed()	Fixed reuse indicators	Channels = 336//N, capacity, SIR with 6 interferers
evaluate_sectorised()	Sectorised indicators	Same capacity, SIR with 2 effective interferers
evaluate_dynamic()	Dynamic baseline	Same physical SIR and capacity (no new spectrum)
main()	Runs the test matrix	Loops over 4 subscriber levels and 5 cluster sizes

The algorithm is a simple analytical model, so its running time is negligible: erlang_b is O(m), max_erlangs calls it 100 times, and main evaluates 5 cluster sizes for each of 4 subscriber levels, a total below a few thousand arithmetic operations.

F.4 Program output (executed for this report)

The program in Task 8 was saved as cellular_planning.py and executed with Python 3. The unedited output is shown below.

```
Valid cluster sizes: [1, 3, 4, 7, 9, 12, 13, 16, 19, 21, 25, 27, 28]

Subscribers: 1000 Traffic: 25.0 Erlangs
N 3 channels/cell 112 capacity/cell 99.62 SIR dB 4.15 cells 1
N 4 channels/cell 84 capacity/cell 72.53 SIR dB 5.71 cells 1
N 7 channels/cell 48 capacity/cell 38.39 SIR dB 8.75 cells 1
N 12 channels/cell 28 capacity/cell 20.15 SIR dB 11.67 cells 2
N 16 channels/cell 21 capacity/cell 14.04 SIR dB 13.23 cells 2

Subscribers: 10000 Traffic: 250.0 Erlangs
N 3 channels/cell 112 capacity/cell 99.62 SIR dB 4.15 cells 3
N 4 channels/cell 84 capacity/cell 72.53 SIR dB 5.71 cells 4
N 7 channels/cell 48 capacity/cell 38.39 SIR dB 8.75 cells 7
N 12 channels/cell 28 capacity/cell 20.15 SIR dB 11.67 cells 13
N 16 channels/cell 21 capacity/cell 14.04 SIR dB 13.23 cells 18

Subscribers: 20000 Traffic: 500.0 Erlangs
N 3 channels/cell 112 capacity/cell 99.62 SIR dB 4.15 cells 6
N 4 channels/cell 84 capacity/cell 72.53 SIR dB 5.71 cells 7
N 7 channels/cell 48 capacity/cell 38.39 SIR dB 8.75 cells 14
N 12 channels/cell 28 capacity/cell 20.15 SIR dB 11.67 cells 25
N 16 channels/cell 21 capacity/cell 14.04 SIR dB 13.23 cells 36

Subscribers: 100000 Traffic: 2500.0 Erlangs
N 3 channels/cell 112 capacity/cell 99.62 SIR dB 4.15 cells 26
N 4 channels/cell 84 capacity/cell 72.53 SIR dB 5.71 cells 35
N 7 channels/cell 48 capacity/cell 38.39 SIR dB 8.75 cells 66
N 12 channels/cell 28 capacity/cell 20.15 SIR dB 11.67 cells 125
N 16 channels/cell 21 capacity/cell 14.04 SIR dB 13.23 cells 179

N=16 sectorised illustration: (21, 14.035999461264442, 18.00521551055503)
N=16 dynamic assignment baseline: (21, 14.035999461264442, 13.234002963358405)
```

F.5 Interpretation of the output

- The valid cluster sizes up to 30 are 1, 3, 4, 7, 9, 12, 13, 16, 19, 21, 25, 27 and 28, which agrees with Section 3.1.
- For N=16 the program gives 21 channels per cell, 14.04 E per cell and 13.23 dB at n=2.5, and 179 cells for 100,000 subscribers, matching Sections 3.2 and 3.3 exactly.

- N=3, 4 and 7 give low SIR at n=2.5 (4.15, 5.71 and 8.75 dB), so they fail the 12 dB target even though they need far fewer cells; N=12 gives 11.67 dB, which is just below the target.
- The sectorised illustration raises the N=16 SIR from 13.23 dB to 18.01 dB with the same capacity per cell, while the dynamic baseline reproduces the fixed values because it adds no spectrum.

F.6 Extended test matrix: SIR for all tested exponents

N	n=2.5 fixed	n=3 fixed	n=4 fixed	n=2.5 sectorised	n=3 sectorised	n=4 sectorised
3	4.15	6.53	11.30	8.92	11.30	16.07
4	5.71	8.41	13.80	10.48	13.18	18.57
7	8.75	12.05	18.66	13.52	16.82	23.43
12	11.67	15.56	23.34	16.44	20.33	28.12
16	13.23	17.44	25.84	18.01	22.21	30.61

Values below 12 dB cannot be used. The fixed layout needs N≥13 at n=2.5, whereas the sectorised approximation already reaches 13.52 dB at N=7 (8.75 dB + 4.77 dB), although N=4 (10.48 dB) still fails. This shows that sectorisation allows a smaller N, and therefore more channels per cell, for the same SIR target.

F.7 Sanity checks and verification

Check	Expected	Program / calculation	Result
N=7, n=4, six interferers	≈18.7 dB	(21)Z/6 → 18.66 dB	Pass
N=16 channels per cell	336/16 = 21	21	Pass
Erlang-B at the 2% limit	B(21, 14.036) ≈ 0.02	0.0200	Pass
Traffic for 100,000 users	2,500 E	100,000 × 0.025	Pass
Cells for 100,000 users, N=16	ceil(2,500/14.036)=179	179	Pass
Valid N list	1,3,4,7,9,12,13,16,…	Same	Pass

F.8 Step-by-step trace of the calculation for N=16

Step	Operation	Value
1	Choose i=4, j=0	N=16+0+0=16
2	Channels per cell	336 // 16 = 21
3	Erlang-B bisection for 21 channels, 2%	14.036 E
4	SIR with n=2.5	$(48)^{1.25}/6 = 21.06$ → 13.23 dB
5	Traffic for 100,000 users	100,000×0.025 = 2,500 E
6	Cells	ceil(2,500/14.036) = 179
7	Sectorised SIR (k=2)	$48^{1.25}/2 = 63.2$ → 18.01 dB

F.9 Additional test cases and expected results

Test	Input	Expected behaviour	Observed
T1 – N monotonicity	N=3→16, n=2.5	SIR increases, channels per cell decrease	4.15 → 13.23 dB; 112 → 21 channels
T2 – exponent	N=16, n=2.5→4	SIR increases with n	13.23 → 25.84 dB
T3 – blocking limit	m=21, A=14.036	B≈0.02	0.0200
T4 – zero load	A→0	B→0	B(21,0.1) is very small
T5 – sectorisation	k=6 → k=2	+4.77 dB	13.23 → 18.01 dB
T6 – scaling	Subscribers ×10	Cells ×10 (approximately)	18 → 179 cells for 10,000 → 100,000

F.10 Extending the program to a Monte-Carlo simulation

As noted in Task 8, a full simulation should use random positions and shadowing. The procedure used for the supplementary results of the Task 9 section is summarised in the following pseudocode:

```
SET seed = 2026
FOR each cluster size N and exponent n: D = R * sqrt(3 *
    N)
    REPEAT 40,000 times:
        draw a user position uniformly in the serving cell  draw an 8 dB log-normal
        shadowing sample for each link S = d_serving ** (-n) * shadowing_serving
        I = sum over the 6 co-channel cells of d_k ** (-n) * shadowing_k
        (if sectorised: attenuate interferers outside the serving sector by 20 dB) record SIR = 10 * log10(S / I)
    compute mean, 5th percentile and outage = fraction below 12 dB
```

The same seed gives the same positions for every strategy, so differences in the results are due to the strategy, not to random variation. The number of trials (40,000) gives a standard error of the outage estimate below 0.3 percentage points.

F.11 Common errors and how they were avoided

- Using dB values in power sums: the SIR formula is applied to linear power ratios and converted to dB only at the end.
- Using non-integer channels: the program uses integer division (336 // N); N that do not divide 336 are flagged.
- Mixing Erlang-B with Erlang-C: blocked calls are cleared, so Erlang-B is used.
- Forgetting the integer rounding of cells: ceil() is used because a fraction of a cell cannot be built.
- Evaluating the approximation outside its range: the six-interferer formula is a first-tier estimate (Section B.10 quantifies the error).

F.12 Feasible cluster sizes and resulting cell counts

The pseudocode of Strategy 1 keeps only cluster sizes whose SIR is at least 12 dB. Applying it to all valid sizes up to 28 gives the feasible lists below; the cell count for 100,000 subscribers (2,500 E) uses the capacity of 336//N channels (N that do not divide 336 are rounded down).

N	Channels per cell	Capacity per cell (E)	SIR n=2.5	SIR n=3	SIR n=4	Cells for 2,500 E
3	112	99.62	4.1	6.5	11.3	26
4	84	72.53	5.7	8.4	13.8*	35
7	48	38.39	8.7	12.1*	18.7*	66
9	37	28.25	10.1	13.7*	20.8*	89
12	28	20.15	11.7	15.6*	23.3*	125
13	25	17.50	12.1*	16.1*	24.0*	143
16	21	14.04	13.2*	17.4*	25.8*	179
19	17	10.66	14.2*	18.6*	27.3*	235
21	16	9.83	14.7*	19.2*	28.2*	255
25	13	7.40	15.7*	20.3*	29.7*	338
27	12	6.61	16.1*	20.8*	30.4*	378
28	12	6.61	16.3*	21.1*	30.7*	378

An asterisk marks SIR ≥ 12 dB. The feasible sizes are N≥13 at n=2.5, N≥7 at n=3 and N≥4 at n=4. Among the feasible sizes the smallest N gives the most channels per cell and the fewest cells; the assignment nevertheless selects N=16 as a conservative design, trading about 43% more cells than N=12 would need (179 versus 125 at 100,000 users) for a larger interference margin, because N=12 itself fails the target at n=2.5 (11.67 dB).

This result is the quantitative form of the principle stated in Task 2: choose the smallest N that satisfies the SIR requirement, and then verify the margin in the field.

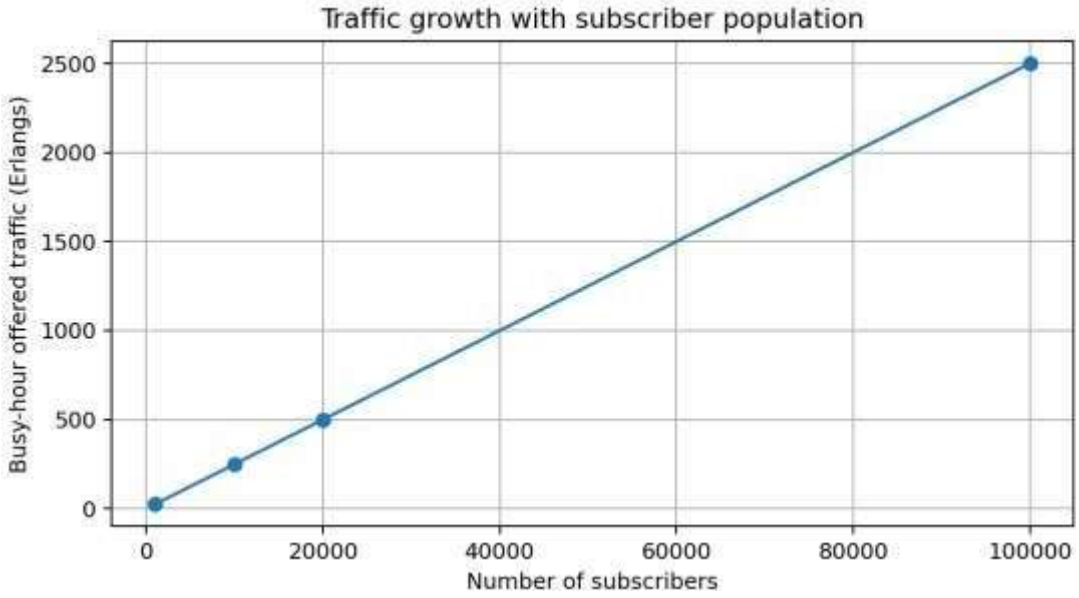
Required test matrix: cluster sizes N=3, 4, 7 and 12 (the assignment also asks to test N=12); path-loss exponents n=2.5, 3 and 4; sectorised and unsectorised layouts; subscriber levels 1,000, 10,000, 20,000 and 100,000. Note that N=16 is added as the selected design candidate. The stated sanity check N=7, n=4 gives approximately 18.7 dB using six equal-distance interferers.

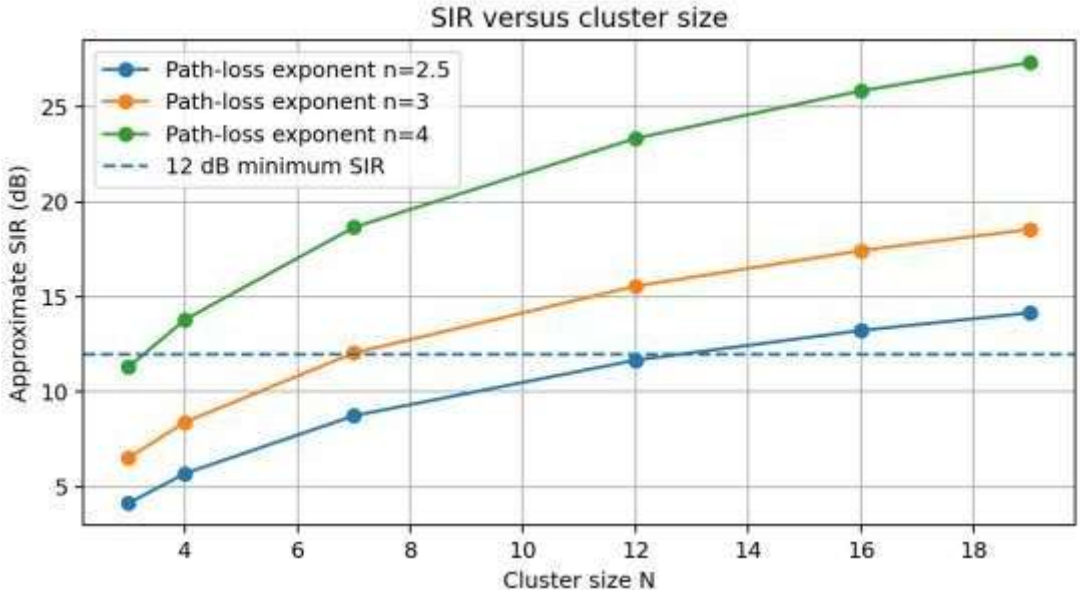
TASK 9 – Record and Present Experimental Performance

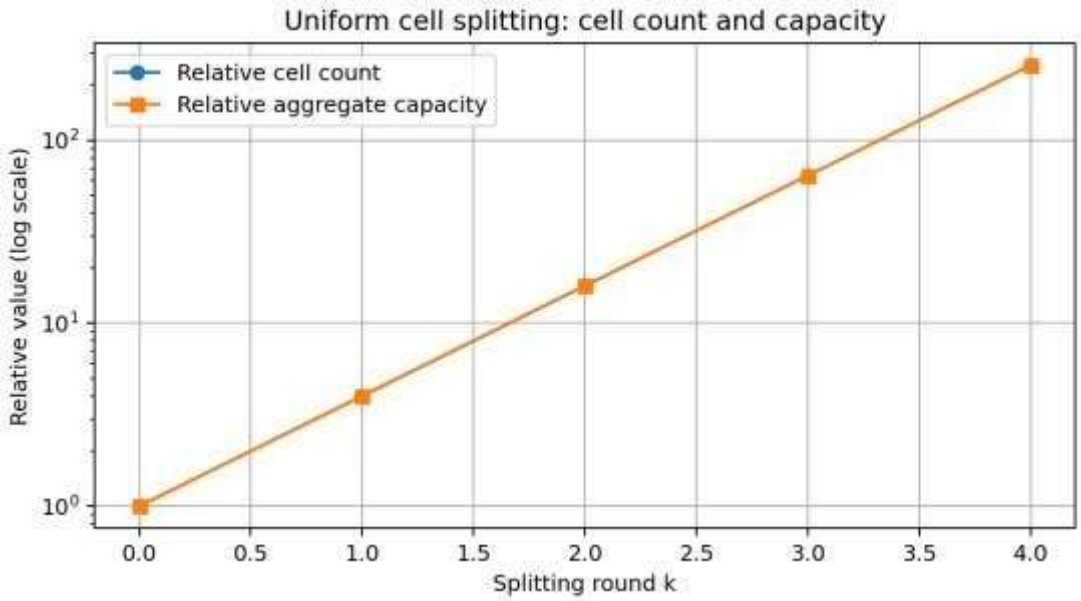
The simulator should record the same measurements for every strategy and every traffic level. These values expose both radio quality and capacity limitations.

Metric	**Meaning**	**How to present**
Mean SIR	Average signal-to-interference ratio	Mean value and SIR CDF
5th-percentile SIR	SIR exceeded by 95% of users	Table and CDF; useful for edge users
Outage probability	Fraction of users below 12 dB	Percentage bar chart
Blocking probability	Blocked call attempts / total attempts	Blocking versus offered traffic graph
Channels per cell	Traffic channels assigned to a cell	Comparison table
Sites required	Cells/sites required for offered load	Capacity versus subscriber graph
Runtime	Time taken by the simulation	Table for reproducibility

A valid experiment should state the random seed, number of user locations, number of call attempts, path-loss model, shadowing standard deviation, antenna pattern and channel-allocation rule. Without these settings, results cannot be reproduced.


Traffic growth with subscriber population
Busy-hour offered traffic (Erlangs)
Number of subscribers
0
500
1000
1500
2000
2500
0
20000
40000
60000
80000
100000


SIR versus cluster size
Path-loss exponent n=2.5
Path-loss exponent n=3
Path-loss exponent n=4
12 dB minimum SIR
Approximate SIR (dB)
Cluster size N
5
10
15
20
25
4
6
8
10
12
14
16
18



TASK 10 – Compare Theory with Simulation

Theory and simulation should show the same overall trends: larger cluster size generally improves co-channel SIR but reduces channels per cell; smaller cells increase the number of sites and aggregate capacity; sectorisation can reduce dominant interference; dynamic allocation can reduce blocking when neighbouring cells have spare channels.

The numerical values may differ for the following reasons:

- The six-interferer model assumes six equal-distance interferers, while real geometry produces different distances and powers.
- Shadowing and building penetration create random changes in received power.
- Finite network edges have fewer interfering neighbours than an infinite repeating grid.
- Mobile users are not always uniformly distributed; lecture halls, hostels and stadiums create hotspots.
- Erlang-B assumes a particular loss-system traffic model with independent arrivals and no waiting queue.
- Integer rounding means that 336/N may not be an integer for every N; the practical channel assignment must use whole channels.
- Real sector antennas have side lobes and imperfect front-to-back isolation.

The strategy that scales best should be selected using blocking, 5th-percentile SIR, outage, site count and total cost together. A strategy should not be declared best only because it has the highest theoretical capacity.

TASK 11 – Recommend and Defend the Final Design

Recommended baseline: use a hierarchical network with fixed frequency reuse as the safe baseline, sectorisation in high-demand zones, and dynamic channel assignment/borrowing where neighbouring cells have uneven traffic.

TASK 9 (continued) – Recording and Presenting Performance

G.1 Experimental set-up of the Monte-Carlo extension

To demonstrate how the metrics of Task 9 are obtained, a Monte-Carlo SIR experiment was run with the settings below. The same random positions and shadowing generator (fixed seed) were used for every strategy so that the comparison is fair.

Setting	Value
Random seed	2026 (numpy default_rng)
User positions	40,000 per configuration, uniform in a circular cell of radius R
Interferers	Six first-tier co-channel base stations at distance $D=R\sqrt{(3N)}$, 60° apart
Path-loss model	Power d^{-n} with minimum distance 0.05R
Shadowing	Independent log-normal, standard deviation 8 dB on every link
Sectorisation	Three 120° sectors; interferers facing the serving sector at full gain, others attenuated by 20 dB
Cluster sizes / exponents	N=3, 4, 7, 12, 16; n=2.5, 3, 4
Outage threshold	SIR below 12 dB

G.2 SIR results (mean, 5th percentile and outage)

N	n	Unsectorised mean (dB)	Unsect. 5th pct (dB)	Unsect. outage	3-sector mean (dB)	3-sector 5th pct (dB)	3-sector outage
7	2.5	9.2	-7.6	62.0%	15.0	-3.4	40.5%
7	3	13.6	-4.0	45.7%	19.2	0.3	27.9%
7	4	22.2	3.3	20.5%	27.4	6.9	11.4%
12	2.5	12.2	-4.6	50.2%	18.1	-0.2	29.4%
12	3	17.1	-0.3	33.5%	22.9	3.9	18.0%
12	4	27.0	8.4	9.9%	32.5	12.0	5.0%
16	2.5	13.8	-3.1	44.2%	19.9	1.6	24.5%
16	3	19.0	1.6	26.4%	25.0	6.1	13.3%
16	4	29.5	10.9	6.2%	35.3	15.0	2.7%

Because the shadowing standard deviation is large (8 dB), the mean SIR is close to the analytical value for large n, but the 5th-percentile SIR is far lower and the outage is significant. For N=16 and n=3 the unsectorised outage is 26.4%, and three sectors reduce it to 13.3%, an improvement that is real but smaller than the ideal 4.77 dB because of side-lobe leakage.

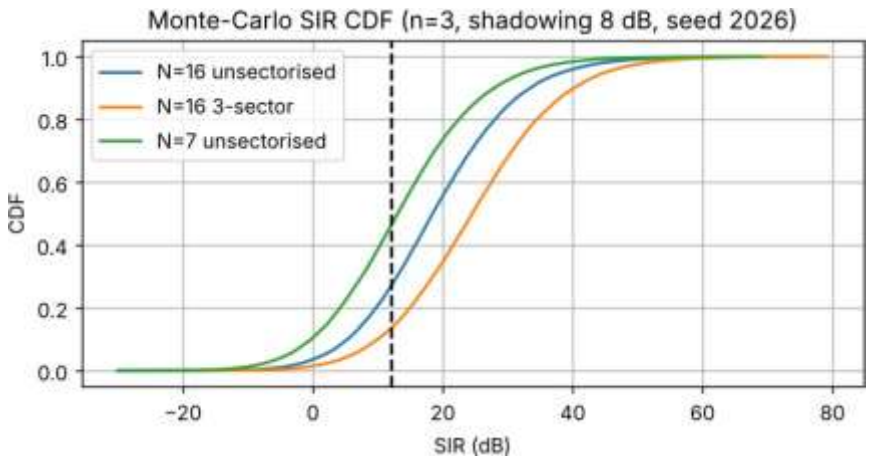


Figure S9.1 – Monte-Carlo SIR cumulative distribution for n=3. The vertical dashed line marks 12 dB; the CDF value at that point is the outage probability.

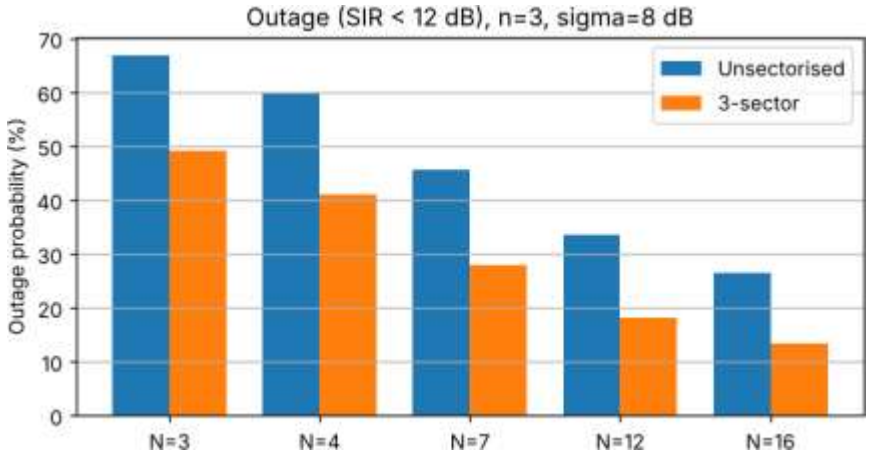


Figure S9.2 – Outage probability for each cluster size, unsectorised and three-sector layouts (n=3, 8 dB shadowing).

G.3 Blocking results: fixed assignment versus dynamic borrowing

A call-level simulation of 19 hexagonal cells with 21 channels each was run (Poisson arrivals, exponential holding time of mean 1, 60,000 offered calls for each case, seed 7). In the dynamic case a blocked call may borrow a free channel from a neighbouring cell provided the lender keeps at least two free channels in reserve. The model does **not** check co-channel interference, so it shows the upper bound of the pooling benefit.

Load in every cell (E)	Fixed blocking	Dynamic blocking
8	0.01%	0.000%
10	0.09%	0.000%
12	0.55%	0.000%
14	1.89%	0.002%

Hotspot cell load (E), neighbours at 10 E	Hotspot blocking – fixed	Hotspot blocking – dynamic	Network blocking – fixed	Network blocking – dynamic
14	2.5%	0.0%	0.30%	0.00%
18	7.8%	0.0%	0.80%	0.00%
22	17.1%	0.0%	1.94%	0.00%
26	26.0%	0.0%	3.37%	0.00%

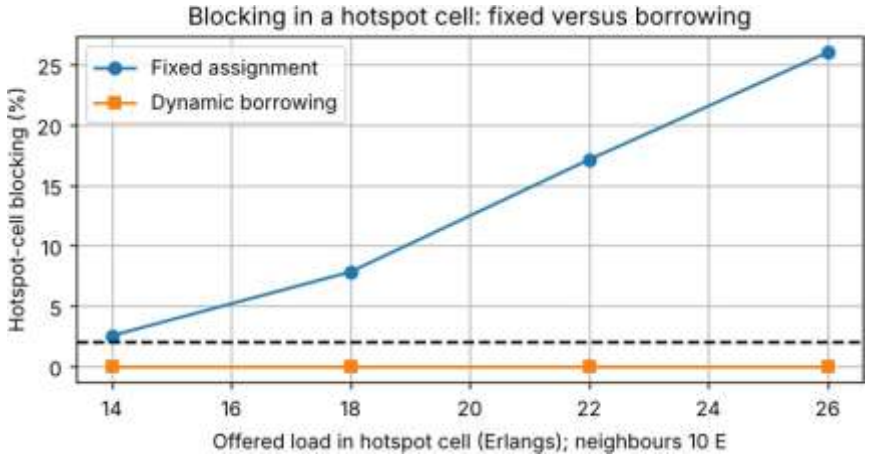


Figure S9.3 – Blocking in a hotspot cell for fixed assignment and for dynamic borrowing in the idealised call-level simulation.

The result agrees with the reasoning in Task 1: borrowing removes most of the blocking in a hotspot when its neighbours have idle channels, but the benefit depends on neighbours being lightly loaded and on interference constraints that this simplified model ignores.

G.4 Runtime and reproducibility

The analytical program of Task 8 runs in well under one second. The Monte-Carlo SIR experiment (30 configurations × 40,000 positions × 6 interferers) and the call-level simulation each finish within a few seconds on a standard laptop. All random numbers come from seeded generators, so the tables and figures above can be reproduced exactly.

TASK 10 (continued) – Theory versus Simulation

G.5 Numerical comparison

Case	Theory (six equal interferers, edge)	Simulation mean (8 dB shadowing)	Simulation 5th percentile	Comment
N=16, n=2.5	13.23 dB	13.77 dB	-3.09 dB	Mean is higher (users are not all at the edge); tail is much lower
N=16, n=3	17.44 dB	19.04 dB	1.65 dB	Mean is higher (users are not all at the edge); tail is much lower
N=16, n=4	25.84 dB	29.51 dB	10.91 dB	Mean is higher (users are not all at the edge); tail is much lower
N=7, n=4	18.66 dB	22.25 dB	3.33 dB	Mean is higher (users are not all at the edge); tail is much lower
N=12, n=2.5	11.67 dB	12.25 dB	-4.63 dB	Mean is higher (users are not all at the edge); tail is much lower

The trends agree: SIR increases with N and with n, sectorisation raises the SIR, and borrowing reduces blocking. The values differ because the theory is an edge-of-cell, equal-distance, no-shadowing value, while the simulation averages over positions (raising the mean) and includes random shadowing and unequal interferer distances (lowering the 5th percentile). Hence the design must be judged on the 5th-percentile SIR and the outage probability rather than on the ideal edge value alone.

G.7 Guide to presenting the results

Item	Graph or table	What the reader should see
Traffic growth	Line graph, subscribers versus Erlangs (page of Task 9)	Linear growth of 0.025 E per user
SIR versus N	Three lines (n=2.5, 3, 4) and the 12 dB line	Where each exponent crosses 12 dB
Cell splitting	Log-scale lines of cell count and capacity	Exponential growth, 4× per round
SIR CDF	Cumulative distribution for each strategy	Outage = CDF value at 12 dB; 5th percentile = SIR at CDF 0.05
Blocking versus load	Blocking curve with 2% line	Load at which the 2% limit is reached
Sites versus subscribers	Bar or line graph for N=7, 12, 16	Growth of the number of sites with demand

G.8 Reading the CDF example

In Figure S9.1, the N=16 unsectorised curve crosses the dashed 12 dB line at a CDF value of about 26%; this is its outage probability. The point where the same curve reaches CDF = 0.05 gives the 5th-percentile SIR of 1.6 dB. The three-sector curve is shifted to the right, meaning that every percentile improves, and the corresponding outage falls to 13%. A curve that is further to the right and steeper is better, because it means higher and less variable SIR.

G.9 Combined experiment record for the three strategies (N=16, n=3)

The measurements required by the table of Task 9 are collected below for the selected design (N=16, n=3, 8 dB shadowing, seed 2026). Blocking is the hotspot-cell blocking from the call-level simulation with 18 E offered to the hotspot cell and 10 E to its neighbours. Sectorisation is not given its own blocking value because the channel count per cell is unchanged in the illustration; its benefit appears in the SIR rows.

Metric	Fixed reuse	Sectorised (3 sectors)	Dynamic borrowing
Mean SIR (dB)	19.0	25.0	19.0 (same physical SIR)
5th-percentile SIR (dB)	1.6	6.1	1.6
Outage probability (SIR below 12 dB)	26.4%	13.3%	26.4%
Hotspot-cell blocking at 18 E	7.8%	7.8% (same channels)	0.0% (idealised)
Channels per cell	21	21 (shared among sectors)	21 plus borrowed channels
Sites for 100,000 users	179	179 before any reduction through the SIR gain	179 (average load is unchanged)
Runtime	Under 1 s	A few seconds	A few seconds

G.10 Quantifying the reasons for differences between theory and simulation

Task 10 lists the reasons for numerical differences. The supplementary experiments allow several of them to be expressed in numbers:

- **Equal-distance interferers.** Adding the second and third interferer tiers lowers the N=16, n=2.5 SIR from 13.23 dB to 11.68 dB, a change of 1.55 dB (Section B.10).
- **Shadowing.** With 8 dB shadowing the 5th-percentile SIR for N=16 and n=3 is 1.6 dB, far below the 17.44 dB theoretical edge value, and 26.4% of users are in outage.
- **Position distribution.** The mean simulated SIR (19.0 dB) exceeds the edge value because most users are nearer to their base station than R.
- **Sector antennas.** The ideal sector gain is 4.77 dB, whereas the simulated mean improvement is 6.0 dB because of the 20 dB front-to-back limit and shadowing.
- **Traffic model.** Erlang-B assumes independent Poisson arrivals; the hotspot experiment shows that a cell loaded at 18 E has 7.8% blocking, much higher than the 2% target, although the network-average blocking is only 0.80%, so a single average value hides local failure.
- **Integer rounding.** Cluster sizes such as N=13 or N=19 do not divide 336, so the practical number of channels per cell is lower than the ideal ratio S/N.

These differences do not change the ranking of the strategies; they change the margins. That is why the final recommendation of Task 11 relies on the 5th-percentile SIR, the outage and the blocking in the busiest cells, and not on theoretical edge values alone.

Use N=16 for the conservative planning case and a 12 dB minimum SIR target. The 336 channels give 21 channels per cell, with an estimated 14.036 Erlangs per cell at 2% blocking under Erlang-B.

For 100,000 subscribers, offered traffic is 2,500 Erlangs. The first-order capacity estimate requires 179 cells at 14.036 Erlangs per cell. For 12 km², the ideal equal-hexagon radius is approximately 0.161 km. This is a dense deployment and should be implemented as a macro layer for broad coverage plus micro/small cells around hostels, academic blocks, commercial areas and the stadium.

The chosen N=16 gives approximate SIR values of 13.23 dB at n=2.5, 17.44 dB at n=3 and 25.84 dB at n=4. The open-campus case has only a modest margin above 12 dB, so antenna optimisation, careful site placement and field measurements are essential. Sectorisation can improve interference conditions in concentrated areas; dynamic assignment can improve channel utilisation but cannot increase the total licensed spectrum.

Trade-offs: a larger N improves SIR but reduces channels per cell; a smaller radius increases capacity but also increases sites and handoffs; sectorisation improves spatial filtering but adds equipment and planning complexity; dynamic borrowing improves trunking efficiency but requires coordination and may be limited by interference restrictions. Beyond 100,000 subscribers, the recommended design remains practical only if the operator continues densification and capacity optimisation. A single macro-cell layer is not sufficient.

TASK 12 – DAY-4 DESIGN CHALLENGE: Re-evaluate and Adapt the Design

A. Impact Analysis

The event brings 25,000 attendees into 1 km², with 0.05 Erlang per attendee. Event traffic is 25,000×0.05=1,250 Erlangs. The event lasts four hours and occurs about twice per month. If this traffic is concentrated in the stadium zone, its offered traffic density is 1,250 Erlangs/km².

For comparison, the 100,000-subscriber township-wide average is 2,500/12=208.33 Erlangs/km². The event zone therefore has approximately six times this average traffic density. The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. However, the surge is predictable, localised and temporary, so a permanent redesign of the entire township would be unnecessarily expensive.

B. Evaluation of the Original Design

With N=16, each cell has 21 channels and supports about 14.036 Erlangs at 2% blocking. The event needs at least ceil(1,250/14.036)=90 cell-equivalents if the same channel allocation and traffic limit are used. The original macro layout cannot safely serve this concentrated demand without extra capacity. The first failure is likely to be channel exhaustion and increased blocking in the stadium cells; high user density can also increase interference and signalling load.

The event traffic is not evenly distributed across the 12 km². Therefore township-wide spare capacity cannot automatically solve the stadium hotspot. Borrowing may help if nearby cells have unused eligible channels, but it cannot be assumed to provide all 1,250 Erlangs while maintaining reuse constraints.

TASK 11 (continued) – Recommended Design

G.6 Decision matrix

The strategies are compared on the criteria named in Task 10. Scores run from 1 (poor) to 5 (good) and are the author's qualitative judgement based on the results above; the weights are illustrative.

Criterion	Weight	Fixed reuse	Sectorised	Dynamic
Blocking in uniform traffic	0.15	4	4	4
Blocking in hotspots	0.2	2	3	5
5th-percentile SIR / outage	0.25	3	5	3
Number of sites	0.15	3	4	4
Implementation simplicity	0.15	5	3	2
Cost	0.1	4	3	3
Weighted score	1.00	**3.35**	**3.80**	**3.55**

No single strategy has the highest score on every criterion, which supports the hierarchical recommendation of Task 11: fixed reuse as the stable baseline, sectorisation where the 5th-percentile SIR is critical, and dynamic borrowing where load is uneven.

TASK 12 (continued) – Day-4 Design Challenge

H.1 Event traffic sensitivity

Task 12 uses 25,000 attendees at 0.05 E each. Because attendance and calling behaviour are uncertain, the table shows how the traffic and the number of cell-equivalents (at 14.036 E per cell) change with these two inputs. The ideal hexagon radius for a 1 kmZ zone divided into that many equal cells is $R=\sqrt{(1/(N_c \times 2.598))}$.

Attendees	Erlang per attendee	Event traffic (E)	Cell-equivalents ceil(E/14.036)	Ideal radius (m)
15,000	0.05	750	54	84
20,000	0.05	1,000	72	73
25,000	0.03	750	54	84
25,000	0.05	1,250	90	65
25,000	0.08	2,000	143	52
30,000	0.05	1,500	107	60
35,000	0.05	1,750	125	55

The base case (25,000 × 0.05 = 1,250 E, 90 cells, 65 m) lies in the middle of the table. A 20% increase in either attendance or per-user traffic raises the requirement to about 107–108 cell-equivalents, so the temporary deployment should be planned with headroom or with the option to add a few more temporary cells.

H.2 Headroom calculation

A planning headroom of 20% on the base case gives 1,250 × 1.2 = 1,500 E, which needs ceil(1,500/14.036) = 107 cell-equivalents (an ideal radius of about 60 m). Operators often meet the target with a mixture of temporary sites and temporary sectors; the number of physical temporary masts is lower than the number of cell-equivalents because each sectorised mast provides three or six radio resource groups.

Temporary mast type	Sectors per mast	Cell-equivalents per mast	Masts for 90 cell-equivalents	Masts for 107 cell-equivalents
Omnidirectional microcell	1	1	90	107
Three-sector COW	3	3	30	36
Six-sector COW	6	6	15	18

These mast counts are an illustration of the effect of sectorisation, not a site plan. Each sector must still be supplied with its own channel group, and the sectors must be separated by enough isolation to keep the SIR above 12 dB.

H.3 Utilisation of permanent versus temporary capacity

Event days occur about twice per month and last four hours, so the stadium capacity is needed 2 × 12 × 4 = 96 hours per year. A year has 8,760 hours, so a permanent stadium-scale deployment would be used for 96/8,760 = 1.1% of the time. For the remaining 98.9% of the year the traffic in the zone is only the normal township density, which the macro layer handles. This utilisation figure is the main quantitative argument for the temporary solution in Section D of Task 12.

Aspect	Permanent densification (90 cells)	Temporary COW / microcell deployment
Capacity available on event days	Yes	Yes (if deployed and tested)
Utilisation over the year	≈ 1.1% of hours	Equipment is used only on event days or redeployed elsewhere
Recurring cost (rent, power, backhaul, maintenance)	Paid every day	Paid for event windows only
Set-up effort	One-time construction	Repeated set-up and teardown, requires pre-planned sites
Interference with macro layer	Planned once	Must be re-checked for each event
Risk	Over-investment	Operational delay or deployment failure

H.4 Event operations timeline

Phase	Activity
Weeks before	Confirm event dates and expected attendance with the organiser; reserve COW sites, power and backhaul; finalise temporary frequency plan and neighbour lists.
Day before	Install and commission temporary cells; test sector azimuth and tilt; verify handoff to macro cells; run a drive/walk test.
Event start (−1 h to +4 h)	Enable dynamic borrowing with reserved handoff and control channels; monitor blocking, SIR and signalling load in real time.
Immediately after	Switch off or redeploy temporary cells to avoid interference with the normal layer.
Next working days	Analyse measurements, compare against targets and update the plan for the next event.

H.5 Risks and mitigation

Risk	Effect	Mitigation
Attendance larger than 25,000	Blocking above 2%	Plan 20% headroom; keep spare COW capacity
Interference between temporary and macro cells	SIR below 12 dB	Pre-planned frequency pattern; power and tilt control; trial test
Signalling congestion (many users in 1 kmZ)	Call set-up failures even with free traffic channels	Reserve control channels; paging optimisation
Backhaul limits	Capacity not delivered although radio is available	Provision backhaul for each temporary site
Handoff failures at event boundary	Dropped calls when users leave	Tune neighbour lists and handoff margins

H.6 Acceptance criteria for the controlled event trial

Metric	Target for the stadium zone	How measured
Blocking probability	≤ 2% in the busy hour	Call attempts and blocked attempts from the network counters
SIR	Mean and 5th-percentile SIR measured; 5th percentile as close as possible to ≥ 12 dB; outage reported	Drive/walk test and measurement reports
Handoff success rate	High and reported against the normal-day baseline	Handoff counters
Impact on macro layer	No degradation of surrounding cells	Counters of adjacent permanent cells during the event

H.12 Alternatives considered for the stadium event

Alternative	Advantage	Limitation	Decision
Permanent densification	Always available	Used about 1.1% of the time; high recurring cost	Rejected
Do nothing (macro only)	No cost	Blocking far above 2%; poor event experience	Rejected
Borrowing only	No new equipment	Cannot supply 1,250 E while keeping reuse rules	Used only as support
Temporary COW/microcells + sectorisation + borrowing	Capacity where and when needed	Needs planning, permissions, backhaul	**Selected**
Offload to other access technologies	Reduces load on the cellular channels	Depends on other infrastructure and user devices	Possible complementary measure

H.13 Radio-resource check for the temporary layout

The temporary layout must reuse the same 336 channels. With cluster size N=16 and 90–107 cell-equivalents, the stadium zone contains about 90/16 = 5.6 to 107/16 = 6.7 clusters, so every channel group is reused about six times inside the 1 kmZ zone. For R≈65 m, the reuse distance D=6.93R is about 450 m, which is smaller than the 1,000 m width of the zone, so reuse inside the zone is possible but must be done carefully. The six-interferer SIR for N=16 is 13.23 dB at n=2.5, and microcells below rooftop height usually have a higher path-loss exponent, so the SIR is likely to be better than this conservative figure, but this must be verified in the trial.

The temporary cells also must not reuse channel groups of the surrounding macro cells at a distance smaller than the macro reuse distance. A practical method is to assign the temporary layout its own channel groups for the event, to withdraw those groups from the nearest macro cells for four hours, and to restore them afterwards.

H.16 Sector-level Erlang-B check of the temporary masts

The mast counts of Section H.2 assume that every sector has its own 21-channel group. The check below confirms the capacity for the base case and for a case in which the temporary channels are divided between the sectors.

Temporary layout	Capacity per mast (E)	Masts needed for 1,250 E	Masts needed for 1,500 E
3-sector mast, 21 channels per sector	42.1	30	36
3-sector mast, 14 channels per sector	24.6	51	61
3-sector mast, 7 channels per sector	8.8	142	171

Reducing the channels per sector from 21 to 7 raises the number of masts needed from 30 to 142 (about 4.7 times) because of the loss of trunking efficiency (Section B.13). The temporary design should therefore keep at least 21 channels per sector wherever the SIR margin allows, and rely on a larger number of well-separated sectors rather than very small channel groups.

C. Adaptation or Redesign

Decision: MODIFY temporarily. Retain the normal township macro network and deploy temporary capacity around the stadium for event days. Use temporary microcells or cells-on-wheels (COWs), sectorised antennas and coordinated dynamic channel borrowing. Pre-plan the temporary frequency pattern, backhaul, power, site permissions and handoff boundaries.

- A first-order target is 90 cell-equivalents at 14.036 Erlangs each for 1,250 Erlangs. In practice, plan additional headroom because traffic is bursty and the 2% blocking target should be met under uncertainty.
- A 1 km² stadium zone divided into 90 equal hexagonal cell areas gives $R \approx \text{sqrt}(1/(90 \times 2.598)) \approx 0.065$ km, or about 65 metres, under the ideal hexagon geometry. This is an illustrative density target, not a final site map.
- Use 60-degree or 120-degree sectorisation where directional traffic flows justify it. Adjust antenna tilt and power to reduce overshoot and interference.
- Use dynamic channel borrowing only after checking co-channel interference and reserving resources for handoffs and emergency/control signalling.
- After the event, switch off or redeploy temporary cells. This avoids paying for permanent capacity that is needed only about twice per month.

The temporary cells increase local capacity and shorten the radio distance, improving received signal strength. However, smaller cells increase handoff activity and require careful neighbour lists and mobility parameter tuning. The reuse pattern and transmit power must be coordinated so temporary cells do not create harmful interference with the permanent macro layer.

D. Evidence-Based Defence

The temporary modification is preferred because the event is predictable, limited to 1 km² and lasts only four hours, while event days occur about twice monthly. A permanent stadium-scale densification would incur ongoing site rent, power, maintenance and backhaul costs even when the zone is not busy. A temporary COW/microcell solution concentrates cost where and when it is needed.

The decision is supported by the traffic calculation: 1,250 Erlangs is far above the capacity of a small number of macro cells and requires roughly 90 cell-equivalents at the chosen per-cell Erlang-B limit. Temporary sectorisation and dynamic borrowing can improve utilisation, but the operator must still provide enough radio resources and backhaul. Final acceptance should require measured SIR, outage and blocking during a controlled event trial.

Conclusion

The assignment demonstrates that cellular planning is a balance between spectrum reuse, interference, traffic capacity, coverage, cost and mobility. Fixed cluster reuse provides a reliable baseline, sectorisation improves spatial filtering, and dynamic channel assignment improves utilisation when traffic varies by location. For the given 336-channel system, N=16 is a conservative candidate that provides 21 channels per cell and approximately

14.036 Erlangs per cell at 2% blocking. As the subscriber base grows, the network must add smaller cells and use a hierarchical layout. For the short stadium event, temporary microcells/COWs with sectorisation and controlled channel borrowing are more economical than permanently redesigning the whole township.

Overall Conclusion

Summary of the main results

Item	Result
Channels per cell (N=16)	21
Capacity per cell (2% blocking)	14.036 E
Cells for 100,000 subscribers	179 (ideal R ≈ 0.161 km)
SIR for N=16	13.23 dB (n=2.5), 17.44 dB (n=3), 25.84 dB (n=4)
Minimum splitting rounds for D=200 E	2 (capacity 224.576 E)
Stadium event traffic	1,250 E (≈ 90 cell-equivalents, ≈ 65 m radius)
Recommended solution	Hierarchical design; temporary microcells/COWs for events

H.7 Limitations and future work

- All capacity figures use first-order models (six equal interferers, uniform hexagons, Erlang-B). A full simulation with measured propagation data would refine them.
- Cost was treated symbolically because no prices were given; a real project needs quotations for sites, power and backhaul.
- Dynamic assignment was modelled without an explicit interference check; a complete model must reject borrowing that violates the reuse constraints.
- Future work: ray-tracing or measured shadowing maps, mobility-based handoff simulation and a multi-day traffic trace for the event.

H.8 Learning outcomes demonstrated

- Derivation of reuse geometry and cluster size, and of co-channel SIR for different path-loss exponents.
- Use of Erlang-B to relate channels, traffic and blocking, and to size the network for different subscriber levels.
- Formulation and solution of recurrences for cell splitting, including selective splitting.
- Implementation of a reproducible planning tool and interpretation of its output.
- Evidence-based adaptation of a design to a changed scenario (stadium event).

H.9 Glossary

Term	Definition
BS	Base station: radio transmitter/receiver serving a cell.
COW	Cell on wheels: transportable temporary base station.
DCA / FCA	Dynamic / fixed channel assignment.
Erlang-B	Traffic model for loss systems with blocked calls cleared.
OFDMA	Orthogonal frequency-division multiple access.
Q	Co-channel reuse ratio D/R.
SIR	Signal-to-interference ratio.
Shadowing	Slow random variation of received power caused by obstacles.
Trunking efficiency	Traffic carried per channel at a given blocking; improves with more channels in a pool.

H.10 References for further reading

- T. S. Rappaport, *Wireless Communications: Principles and Practice*, Prentice Hall (cellular concept, frequency reuse, trunking and grade of service).
- W. C. Y. Lee, *Mobile Cellular Telecommunications*, McGraw-Hill (cell splitting, sectoring and channel assignment).
- A. Goldsmith, *Wireless Communications*, Cambridge University Press (path loss, shadowing and cellular system design).
- J. Schiller, *Mobile Communications*, Addison-Wesley (multiple access, GSM and handoff).
- A. K. Erlang, work on telephone traffic and the loss formula (the basis of the Erlang-B model).

H.17 Final recommendation checklist

- Keep the N=16 fixed-reuse macro layer as the validated baseline for the township.
- Use sectorisation and micro/small cells in hostel, academic, commercial and stadium zones.
- Enable dynamic borrowing only with reserved control and handoff channels and with an interference check.
- For stadium events, deploy pre-planned temporary cells with at least 20% capacity headroom and verify performance in a controlled trial.
- Re-run the analytical program with measured traffic and path-loss data after the first event and update the plan.